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This is a corrected copy of the paper cited above. The text is retyped from the printed paper. It keeps the paper's wording and the numbering of its results, equations and footnotes, and a marker such as [p. $N$] shows where page $N$ of the printed paper begins.

The corrections come from a formalization of the Graph Minors series in the Lean proof assistant, github.com/mahdi-jfri/GraphMinor, in which the results of this paper are proved. Where the formalization found that a statement or a step of a proof is wrong as printed, this copy corrects it in place. Where the paper leaves a proof or a step to the reader, this copy adds it, also in blue. Text ~~struck through in red~~ is removed and text in blue is inserted. Each change has a note, marked F1, F2, ..., that gives the reason. The file corrections/GM10.md of the formalization describes each correction in full. Each note, and each item in the list below, begins with the kind of the change. A **misprint** is a wrong symbol, index or word. A **statement mistake** is a result or a claim of the text that is false as printed. A **degenerate case** is a statement that fails only in a trivial case, such as an empty graph or a parameter equal to 0. A **proof gap** is a step of a proof that does not follow as written. A **missing proof** is a proof that the paper omits, leaves to the reader or only sketches, and that this copy adds.

The corrections and added proofs, in the order of the pages:

1. **Missing proof.** Proof of (2.6), page 156: the details of the sketched proof are added.
2. **Proof gap.** Proof of (4.2)(ii), page 165: a direct argument for "only if", which covers graphs with loops, is added.
3. **Misprint.** Section 7, page 171: in the definition of $\partial(X)$, $v \in X$ becomes $v \in V(G)$.
4. **Missing proof.** Proof of (9.1), page 177: the proof is added.
5. **Missing proof.** Proof of (10.4), page 184: the details of the sketched proof are added.

# [p. 153] Graph Minors. X. Obstructions to Tree-Decomposition

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Roughly, a graph has small "tree-width" if it can be constructed by piecing small graphs together in a tree structure. Here we study the obstructions to the existence of such a tree structure. We find, for instance:

(i) a minimax formula relating tree-width with the largest such obstructions

(ii) an association between such obstructions and large grid minors of the graph

(iii) a "tree-decomposition" of the graph into pieces corresponding with the obstructions.

These results will be of use in later papers.

## 1. Tangles

Graphs in this paper are finite and undirected and may have loops or multiple edges. The vertex- and edge-sets of a graph $G$ are denoted by $V(G)$ and $E(G)$. If $G_1 = (V_1, E_1)$, $G_2 = (V_2, E_2)$ are subgraphs of a graph $G$, we denote the graphs $(V_1 \cap V_2, E_1 \cap E_2)$ and $(V_1 \cup V_2, E_1 \cup E_2)$ by $G_1 \cap G_2$ and $G_1 \cup G_2$, respectively. A *separation* of a graph $G$ is a pair $(G_1, G_2)$ of subgraphs with $G_1 \cup G_2 = G$ and $E(G_1 \cap G_2) = \emptyset$, and the *order* of this separation is $|V(G_1 \cap G_2)|$.

It sometimes happens with a graph $G$ that for each separation $(G_1, G_2)$ of $G$ of low order, we may view one of $G_1, G_2$ as the "main part" of $G$, in [p. 154] a consistent way. For example if $G$ is drawn on a connected surface (not a sphere) and every non-null-homotopic curve in the surface meets the drawing

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many times, then it can be shown (see [5]) that for each low order separation $(G_1, G_2)$, exactly one of $G_1, G_2$ contains a non-null-homotopic circuit. As a second example, let $H$ be a minor of $G$ (defined later), isomorphic to a large complete graph; then for each low order separation $(G_1, G_2)$ of $G$, exactly one of $G_1, G_2$ has a subgraph corresponding to a vertex of $H$. The object of this paper is to study such "tangles," as we call them, since they play a central role in future papers of this series.

Many of our results about tangles extend easily to hypergraphs, and we have expressed them in this generality. A *hypergraph* $G$ consists of a set of vertices $V(G)$, a set of edges $E(G)$, and an incidence relation; each edge may or may not be incident with each vertex. If each edge is incident with either one or two vertices, the hypergraph is a *graph*. All hypergraphs in this paper are finite. A *subhypergraph* $G'$ of $G$ is a hypergraph such that

(i) $V(G') \subseteq V(G)$, $E(G') \subseteq E(G)$

(ii) for $e \in E(G')$ and $v \in V(G)$, $e$ is incident with $v$ in $G$ if and only if $v \in V(G')$ and $e$ is incident with $v$ in $G'$.

We write $G' \subseteq G$ if $G'$ is a subhypergraph of $G$. We define $G_1 \cup G_2$, $G_1 \cap G_2$ for subhypergraphs $G_1, G_2$ of a hypergraph as for graphs, and a separation of a hypergraph, and its order, are defined as for graphs. If $G$ is a hypergraph and $X \subseteq E(G)$, $G \setminus X$ is the subhypergraph $G'$ with $V(G') = V(G)$, $E(G') = E(G) - X$; while if $X \subseteq V(G)$, $G \setminus X$ is the subhypergraph with $V(G') = V(G) - X$ and $E(G')$ the set of those edges of $G$ incident with no vertex in $X$. We sometimes abbreviate $G \setminus \{x\}$ to $G \setminus x$, etc.

Let $G$ be a hypergraph and let $\theta \geq 1$ be an integer. A *tangle* in $G$ of *order* $\theta$ is a set $\mathscr{T}$ of separations of $G$, each of order $< \theta$, such that

(i) for every separation $(A, B)$ of $G$ of order $< \theta$, one of $(A, B)$, $(B, A)$ is in $\mathscr{T}$

(ii) if $(A_1, B_1), (A_2, B_2), (A_3, B_3) \in \mathscr{T}$ then $A_1 \cup A_2 \cup A_3 \neq G$

(iii) if $(A, B) \in \mathscr{T}$ then $V(A) \neq V(G)$.

We refer to these as the *first*, *second*, and *third (tangle) axioms*. Every tangle $\mathscr{T}$ has order $\leq |V(G)|$, since $(G, G \setminus E(G)), (G \setminus E(G), G) \notin \mathscr{T}$. The *tangle number* of $G$, denoted $\theta(G)$, is the maximum order of tangles in $G$ (or 0, if there are no tangles).

The main results of this paper are as follows:

(1) Tangle number is connected with "tree-width," which was discussed in earlier papers of this series (for example, [3]); indeed, there is a [p. 155] minimax equation connecting the tangle number of a hypergraph and its "branch-width," which is an invariant very similar to tree-width and essentially within a constant factor of tree-width.

(2) Despite our rather abstract definition of a tangle, there are in any hypergraph $G$ at most $|V(G)|$ maximal tangles, and any other tangle is a subset (a "truncation") of one of these. Furthermore, there is a "tree-decomposition" of $G$, the vertices of which correspond to these maximal tangles.

(3) For $\theta \geq 2$, any minor isomorphic to a $(\theta \times \theta)$-grid of a graph $G$ gives rise to a tangle in $G$ of order $\theta$, and conversely, for any $\theta \geq 2$ there exists $N(\theta) \geq \theta$ such that for every tangle of order $\geq N(\theta)$ in a graph $G$, its truncation to order $\theta$ is the tangle arising from some $(\theta \times \theta)$-grid minor of $G$.

(4) Finally, the main result of the paper. It is too technical to state without a number of definitions, but roughly it enables us to gain knowledge of the global structure of a hypergraph from a knowledge of its structure relative to each tangle. This will be applied in [6].

## 2. Some Tangle Lemmas

In this section we develop some easy results about tangles for later use.

(2.1) *If $\mathscr{T}$ is a tangle and $(A, B) \in \mathscr{T}$ then $(B, A) \notin \mathscr{T}$.*

*Proof.* Since $A \cup B = G$, $(B, A) \notin \mathscr{T}$ by the second axiom. □

(2.2) *If $\mathscr{T}$ is a tangle of order $\theta$ and $(A, B), (A', B') \in \mathscr{T}$ and $(A \cup A', B \cap B')$ has order $< \theta$ then $(A \cup A', B \cap B') \in \mathscr{T}$.*

*Proof.* Now $(B \cap B', A \cup A') \notin \mathscr{T}$ by the second axiom, because $(A, B), (A', B') \in \mathscr{T}$ and $A \cup A' \cup (B \cap B') = G$. Thus $(A \cup A', B \cap B') \in \mathscr{T}$ by the first axiom. □

(2.3) *If $\mathscr{T}$ has order $\geq 2$ and $(A_1, B_1), (A_2, B_2), (A_3, B_3) \in \mathscr{T}$ then $E(A_1 \cup A_2 \cup A_3) \neq E(G)$.*

*Proof.* Suppose that there exist $(A_1, B_1), (A_2, B_2), (A_3, B_3) \in \mathscr{T}$ with $E(A_1 \cup A_2 \cup A_3) = E(G)$, and choose them with $|V(A_1)|$ maximum. By the second axiom, $A_1 \cup A_2 \cup A_3 \neq G$, and so there is a vertex $v$ of $G$ in none of $V(A_1)$, $V(A_2)$, $V(A_3)$ and hence incident with no edge of $G$. Let $K$ be the hypergraph with $V(K) = \{v\}$, $E(K) = \emptyset$. Then $(K, G)$ has order 1 and by the second axiom, $(G, K) \notin \mathscr{T}$; thus $(K, G) \in \mathscr{T}$ by the first axiom, since $\mathscr{T}$ [p. 156] has order $\geq 2$. Now $(K, G \setminus v)$ has order 0, and $(G \setminus v, K) \notin \mathscr{T}$ by the second axiom, since $(G \setminus v) \cup K = G$. Thus $(K, G \setminus v) \in \mathscr{T}$. But $(K \cup A_1, (G \setminus v) \cap B_1)$ has order at most the order of $(A_1, B_1)$ and hence is in $\mathscr{T}$ by (2.2), contrary to the maximality of $|V(A_1)|$, as required. □

For an edge $e$ of a hypergraph $G$, the *ends* of $e$ are the vertices of $G$ incident with $e$, and the *size* of $e$ is the number of ends of $e$.

(2.4) *Let $\theta \geq 1$, and let $e$ be an edge of $G$ with size $\geq \theta$. Let $\mathscr{T}$ be the set of all separations $(A, B)$ of $G$ of order $< \theta$ with $e \in E(B)$. Then $\mathscr{T}$ is a tangle of order $\theta$.*

*Proof.* The first two axioms are clear. For the third, let $(A, B) \in \mathscr{T}$. Then $V(A \cap B)$ does not contain every end of $e$ since $|V(A \cap B)| < \theta$, and yet $e \in E(B)$, and so $V(A) \neq V(G)$. This completes the proof. □

We remark

(2.5) *$G$ has a tangle if and only if $V(G) \neq \emptyset$.*

*Proof.* If $v \in V(G)$, let $\mathscr{T}$ be the set of all separations $(A, B)$ of $G$ of order 0 with $v \in V(B)$. Then $\mathscr{T}$ is a tangle of order 1, as is easily seen. Conversely, since every tangle has order $\leq |V(G)|$, if $G$ has a tangle then $V(G) \neq \emptyset$. □

For graphs, we can extend (2.5) as follows.

(2.6) *If $G$ is a graph, the tangles in $G$ of order 1 are in 1–1 correspondence with the connected components of $G$, and those of order 2 are in 1-1 correspondence with the blocks of $G$ which have a non-loop edge.*

(A block of a graph is a maximal connected subgraph any two distinct edges of which are in a circuit.)

*Proof.* Since we do not need the result, we merely sketch the proof. Any $v \in V(G)$ yields a tangle of order 1 as in (2.5), and it is easy to see that every tangle of order 1 arises this way, and distinct $v, v' \in V(G)$ yield the same tangle if and only if $v$ and $v'$ are in the same component of $G$. For order 2, any non-loop edge yields a tangle of order 2, by (2.4), and again, it is easy to see that every order 2 tangle arises this way, and two edges yield the same tangle if and only if they are in the same block. We give the details. Besides (2.2)–(2.4) we use (2.7), (2.9) and the extreme separations defined before (2.8), whose proofs do not use (2.6). Let $K$ be the null graph. For every tangle $\mathscr{T}$ in $G$, the pair $(K, G)$ is in $\mathscr{T}$, since $(K, G)$ and $(G, K)$ are separations of order 0 and $(G, K) \notin \mathscr{T}$ by the third axiom.

For $v \in V(G)$ let $\mathscr{T}_v$ be the set of all separations $(A, B)$ of $G$ of order 0 with $v \in V(B)$. This is the tangle of order 1 of (2.5). Indeed, if $(A, B)$ has order 0 then $v$ is in exactly one of $V(A)$, $V(B)$, which gives the first axiom, and if $(A_i, B_i) \in \mathscr{T}_v$ $(1 \leq i \leq 3)$ then $v$ is in none of $V(A_1)$, $V(A_2)$, $V(A_3)$, which gives the second and third.

(1) *Every tangle $\mathscr{T}$ in $G$ of order 1 equals $\mathscr{T}_v$ for some $v \in V(G)$.*

Choose $(A, B) \in \mathscr{T}$ with $|V(A)| + |E(A)| - |V(B)| - |E(B)|$ maximum (this is possible since $(K, G) \in \mathscr{T}$). Then $(A, B)$ is extreme, since for $(A', B') \in \mathscr{T}$ with $A \subseteq A'$ and $B' \subseteq B$ the maximality gives $A' = A$ and $B' = B$. By the third axiom there exists $v \in V(G) - V(A)$, and $v \in V(B)$

since $A \cup B = G$. Let $(A', B') \in \mathscr{T}$. A vertex in both $A \cup A'$ and $B \cap B'$ is in $A \cap B$ or in $A' \cap B'$, and so $(A \cup A', B \cap B')$ has order 0. Hence it is in $\mathscr{T}$ by (2.2), and $B \cap B' = B$ since $(A, B)$ is extreme. Thus $v \in V(B')$, and so $\mathscr{T} \subseteq \mathscr{T}_v$. Conversely, let $(A', B') \in \mathscr{T}_v$. If $(B', A') \in \mathscr{T}$ then $v \in V(A')$ by what we just proved, and so $v \in V(A' \cap B')$, which is impossible since $(A', B')$ has order 0. Hence $(A', B') \in \mathscr{T}$ by the first axiom. This proves (1).

(2) *For $u, v \in V(G)$, $\mathscr{T}_u = \mathscr{T}_v$ if and only if $u$ and $v$ are in the same component of $G$.*

Let $(A, B)$ be a separation of order 0. An edge of $G$ with an end in $V(B)$ is not in $E(A)$, since all ends of an edge of $A$ are in $V(A)$ and no vertex is in both. So it is in $E(B)$, and all its ends are in $V(B)$. Hence if $u$ and $v$ are joined by a path of $G$, then $u \in V(B)$ if and only if $v \in V(B)$, and so $\mathscr{T}_u = \mathscr{T}_v$. Now let $u$ and $v$ be in different components, let $A$ be the component that contains $u$, and let $B$ be the subgraph formed by all the other vertices and edges of $G$. Then $(A, B)$ is a separation of order 0 since no edge of $B$ has an end in $V(A)$, and $(A, B) \in \mathscr{T}_v - \mathscr{T}_u$. This proves (2).

By (1) and (2), mapping each component $C$ of $G$ to $\mathscr{T}_v$, where $v \in V(C)$, gives a 1-1 correspondence between the components of $G$ and the tangles of order 1. (By (2) the image does not depend on the choice of $v$ and distinct components have distinct images, and by (1) every tangle of order 1 is an image.)

For a non-loop edge $e$ let $\mathscr{T}_e$ be the set of all separations $(A, B)$ of $G$ of order $\leq 1$ with $e \in E(B)$. Since $e$ has size 2, $\mathscr{T}_e$ is a tangle of order 2 by (2.4).

(3) *Every tangle $\mathscr{T}$ in $G$ of order 2 equals $\mathscr{T}_e$ for some non-loop edge $e$.*

For $X \subseteq E(G)$, let $L(X)$ be the subgraph with edge set $X$ whose vertices are the ends of the edges in $X$ and the vertices incident with no edge, and let $R(X)$ be the subgraph with edge set $E(G) - X$ whose vertices are the ends of the edges not in $X$. Then $(L(X), R(X))$ is a separation of $G$. If $(A, B) \in \mathscr{T}$ then $(L(E(A)), R(E(A))) \in \mathscr{T}$. For a vertex in both $L(E(A))$ and $R(E(A))$ is an end of an edge of $A$ and of an edge of $B$, and so it is in $V(A \cap B)$. Hence $(L(E(A)), R(E(A)))$ has order $\leq 1$, and it is in $\mathscr{T}$ by (2.9)(iii). In particular $(L(\emptyset), R(\emptyset)) \in \mathscr{T}$, since $(K, G) \in \mathscr{T}$.

We claim that some edge $e$ of $G$ is in $E(B)$ for all $(A, B) \in \mathscr{T}$. Suppose not. Then every edge is in $E(A)$ for some $(A, B) \in \mathscr{T}$, and so it is in some $X \subseteq E(G)$ with $(L(X), R(X)) \in \mathscr{T}$. Choose a set $\mathscr{S}$ of subsets $X$ of $E(G)$ with union $E(G)$, such that $(L(X), R(X)) \in \mathscr{T}$ for all $X \in \mathscr{S}$, with $|\mathscr{S}|$ minimum. If $|\mathscr{S}| \leq 3$, then the members of $\mathscr{S}$ and copies of $\emptyset$ give

three members of $\mathscr{T}$ the union of whose first terms contains every edge of $G$, contrary to (2.3). So $|\mathscr{S}| \geq 4$. Choose $X_1 \in \mathscr{S}$, and put $(A_1, B_1) = (L(X_1), R(X_1))$. Let $X' \in \mathscr{S} - \{X_1\}$, and put $(A', B') = (L(X'), R(X'))$. Then $(A_1 \cup A', B_1 \cap B')$ has order $\geq 2$. For otherwise it is in $\mathscr{T}$ by (2.2), and so $(L(X_1 \cup X'), R(X_1 \cup X')) \in \mathscr{T}$ by the previous paragraph, since $E(A_1 \cup A') = X_1 \cup X'$. But then replacing $X_1$ and $X'$ by $X_1 \cup X'$ in $\mathscr{S}$ gives a smaller set, contrary to the choice of $\mathscr{S}$. Now every vertex of $(A_1 \cup A') \cap (B_1 \cap B')$ is either in $V(A_1 \cap B_1) \cap V(B')$ or in $V(A' \cap B') - V(A_1)$. Each of these two sets has at most one member, since both separations have order $\leq 1$, and together they have at least two. So each has exactly one. Hence $V(A_1 \cap B_1) = \{u\}$ for some $u \in V(B')$, and the only vertex of $A' \cap B'$ is not in $V(A_1)$, so it is not $u$. Since $u \in V(B')$, it follows that $u \notin V(A')$. Here $u$ does not depend on $X'$, since $V(A_1 \cap B_1) = \{u\}$. Now $u \in V(B_1)$, and so $u$ is an end of some edge $f \notin X_1$. Since the union of $\mathscr{S}$ is $E(G)$, $f \in X'$ for some $X' \in \mathscr{S} - \{X_1\}$, and then $u \in V(L(X'))$, a contradiction. This proves our claim.

Choose $e$ as in the claim. If $e$ is a loop, then $e$ has size 1, and so $(K_e, G \setminus e) \in \mathscr{T}$ by (2.7), where $K_e$ is formed by $e$ and its end. This is impossible since $e \notin E(G \setminus e)$. So $e$ is not a loop, and $\mathscr{T} \subseteq \mathscr{T}_e$ by the claim. Conversely, let $(A, B) \in \mathscr{T}_e$. Then $e \notin E(A)$, and so $(B, A) \notin \mathscr{T}$ by the claim. Hence $(A, B) \in \mathscr{T}$ by the first axiom. This proves (3).

(4) *If two distinct edges $e_1, e_2$ of $G$ are in a circuit $C$, then $\mathscr{T}_{e_1} = \mathscr{T}_{e_2}$.*

Since $C$ has two edges, it is not a loop, and so every edge of $C$ has two ends and every vertex of $C$ is incident with exactly two edges of $C$. Let $(A, B)$ be a separation of order $\leq 1$. For $v \in V(C)$ let $d(v)$ be the number of edges of $C$ in $E(A)$ incident with $v$. The sum of $d(v)$ over all $v \in V(C)$ is twice the number of edges of $C$ in $E(A)$, and so it is even. If $v \in V(A) - V(B)$ then $d(v) = 2$, since an edge of $B$ at $v$ would put $v$ in $V(B)$, and if $v \notin V(A)$ then $d(v) = 0$. Since at most one vertex of $C$ is in $V(A \cap B)$, it follows that $d(v)$ is even for that vertex too. So at each vertex of $C$, either both edges of $C$ are in $E(A)$ or neither is. Since $C$ is connected, either every edge of $C$ is in $E(A)$ or none is. Since every edge of $G$ is in exactly one of $E(A)$, $E(B)$, it follows that $e_1 \in E(B)$ if and only if $e_2 \in E(B)$. This proves (4).

(5) *If $e_1, e_2$ are distinct non-loop edges with $\mathscr{T}_{e_1} = \mathscr{T}_{e_2}$, then some circuit of $G$ contains both.*

Let $Y_1$, $Y_2$ be the sets of ends of $e_1$, $e_2$, and let $G' = G \setminus \{e_1, e_2\}$. Suppose that $G'$ has a separation $(G_1, G_2)$ of order $\leq 1$ with $Y_1 \subseteq V(G_1)$ and $Y_2 \subseteq V(G_2)$. Adding $e_1$ to $G_1$ and $e_2$ to $G_2$ gives a separation $(A, B)$ of $G$ of order $\leq 1$ with $e_1 \in E(A)$ and $e_2 \in E(B)$. Then $(A, B) \in \mathscr{T}_{e_2} = \mathscr{T}_{e_1}$, and so $e_1 \in E(B)$, a contradiction. Hence by Menger's theorem there are two

disjoint paths of $G'$ from $Y_1$ to $Y_2$. Since they are disjoint and $|Y_1| = |Y_2| = 2$, one of them joins one end of $e_1$ to one end of $e_2$, the other joins the other two ends, and neither has an internal vertex in $Y_1 \cup Y_2$. So the two paths together with $e_1$ and $e_2$ form a circuit of $G$. This proves (5).

For a non-loop edge $e$, let $H_e$ be the subgraph of $G$ whose edges are the non-loop edges $f$ with $\mathscr{T}_f = \mathscr{T}_e$, and whose vertices are the ends of these edges.

(6) *For every non-loop edge $e$, $H_e$ is a block of $G$, and every block of $G$ that contains $e$ equals $H_e$.*

Let $f_1, f_2$ be distinct edges of $H_e$. Then $\mathscr{T}_{f_1} = \mathscr{T}_{f_2}$, and so some circuit $C$ of $G$ contains $f_1$ and $f_2$, by (5). Every edge of $C$ is a non-loop edge with the same tangle as $f_1$, by (4), and so $C \subseteq H_e$. Hence any two distinct edges of $H_e$ are in a circuit of $H_e$. Also $H_e$ is connected, since any two of its vertices are ends of edges $f_1, f_2$ of $H_e$, and either $f_1 = f_2$ or $f_1, f_2$ are in a circuit of $H_e$. Now let $H$ be a connected subgraph of $G$ that contains $e$, any two distinct edges of which are in a circuit of $H$. We show that $H \subseteq H_e$. Each edge $f \neq e$ of $H$ is in a circuit of $H$ together with $e$. This circuit is not a loop, and so $f$ is a non-loop edge with $\mathscr{T}_f = \mathscr{T}_e$, by (4). So $E(H) \subseteq E(H_e)$. Each vertex of $H$ is incident with an edge of $H$, since $H$ is connected and contains the edge $e$. So $V(H) \subseteq V(H_e)$, and $H \subseteq H_e$. Applying this to a subgraph $H$ that contains $H_e$ shows that $H_e$ is maximal, and so $H_e$ is a block. Applying it to a block $H$ that contains $e$ gives $H \subseteq H_e$, and so $H = H_e$ by the maximality of $H$. This proves (6).

For non-loop edges $e, e'$, $H_e = H_{e'}$ if and only if $\mathscr{T}_e = \mathscr{T}_{e'}$, by the definition of $H_e$ and since $e \in E(H_e)$. Map each block $H$ with a non-loop edge to $\mathscr{T}_e$, where $e$ is a non-loop edge of $H$. By (6), $H = H_e$ for every such $e$, and so the image does not depend on the choice of $e$, and distinct blocks have distinct images. By (3) and (6), every tangle of order 2 is $\mathscr{T}_e$ for some non-loop edge $e$, and so it is the image of the block $H_e$. This gives the 1-1 correspondence for order 2.[F1] □

One might speculate that in a graph, the tangles of order $d$ correspond to the long-sought "$d$-connected components," but that possibility is not further explored here.

[p. 157] Some further lemmas:

(2.7) *Let $\mathscr{T}$ be a set of separations of a hypergraph $G$, each of order $< \theta$, satisfying the first and second tangle axioms. Then $\mathscr{T}$ is a tangle if and only if $(K_e, G \setminus e) \in \mathscr{T}$ for every $e \in E(G)$ of size $< \theta$, where $K_e$ is the*

[F1] **Missing proof.** The paper only sketches this proof. The details above are the argument used in the formalization.

*hypergraph formed by $e$ and its ends.*

*Proof.* If $\mathscr{T}$ is a tangle and $e \in E(G)$ then $(G \setminus e, K_e) \notin \mathscr{T}$ by the third tangle axiom, since $V(G \setminus e) = V(G)$, and so $(K_e, G \setminus e) \in \mathscr{T}$, as required. For the converse, let $\mathscr{T}$ not be a tangle, and choose $(A, B) \in \mathscr{T}$ with $V(A) = V(G)$ and with $B$ minimal. By the second tangle axiom, $A \neq G$ and so $E(B) \neq \emptyset$; choose $e \in E(B)$. From the minimality of $B$, $(A \cup K_e, B \setminus e) \notin \mathscr{T}$, and so $(B \setminus e, A \cup K_e) \in \mathscr{T}$. Hence $(K_e, G \setminus e) \notin \mathscr{T}$ by the second axiom, since $(A, B) \in \mathscr{T}$ and $A \cup (B \setminus e) \cup K_e = G$. But $e$ has size $< \theta$, since every end of $e$ is in $V(A \cap B)$. The result follows. □

Let $\mathscr{T}$ be a tangle in a hypergraph $G$. A separation $(A, B) \in \mathscr{T}$ is *extreme* if $A' = A$ and $B' = B$ for every $(A', B') \in \mathscr{T}$ with $A \subseteq A'$ and $B' \subseteq B$.

(2.8) *Let $\mathscr{T}$ be a tangle of order $\theta$ in a hypergraph $G$, and let $(A, B) \in \mathscr{T}$ be extreme. Then $(A, B)$ has order $\theta - 1$. Moreover, if $(B_1, B_2)$ is a separation of $B$, then either $B_1 \subseteq A \cap B$ and $B_2 = B$, or $B_2 \subseteq A \cap B$ and $B_1 = B$, or $(B_1, B_2)$ has order strictly greater than*

$$\min(|V(A \cap B_1)|, |V(A \cap B_2)|).$$

*In particular, there is no separation $(B_1, B_2)$ of $B$ with $B_1, B_2$ non-null of order 0, and there is no edge of $B$ with all its ends in $V(A)$.*

*Proof.* By the third axiom there exists $v \in V(B) - V(A)$. Let $K_v$ be the hypergraph with vertex set $\{v\}$ and with no edges. From the extremity of $(A, B)$, $(A \cup K_v, B) \notin \mathscr{T}$, and $(B, A \cup K_v) \notin \mathscr{T}$ by the second axiom, since $(A, B) \in \mathscr{T}$ and $A \cup B = G$. Thus $(A \cup K_v, B)$ has order $\geq \theta$, and so $(A, B)$ has order $\theta - 1$.

Let $(B_1, B_2)$ be a separation of $B$. If $(A \cup B_1, B_2) = (A, B)$ then $B_1 \subseteq A$ and $B_2 = B$, and so we may assume that $(A \cup B_1, B_2) \neq (A, B)$. From the extremity of $(A, B)$, $(A \cup B_1, B_2) \notin \mathscr{T}$, and similarly $(A \cup B_2, B_1) \notin \mathscr{T}$. Not both $(B_2, A \cup B_1)$, $(B_1, A \cup B_2) \in \mathscr{T}$, by the second axiom, since $A \cup B_1 \cup B_2 = G$, and without loss of generality we assume that $(B_2, A \cup B_1) \notin \mathscr{T}$. Since $(A \cup B_1, B_2) \notin \mathscr{T}$ it follows that $(A \cup B_1, B_2)$ has order $\geq \theta$; that is,

$$|V(B_1 \cap B_2)| + |V(A \cap B) - V(A \cap B_1)| \geq \theta = |V(A \cap B)| + 1.$$

Hence $|V(B_1 \cap B_2)| > |V(A \cap B_1)|$, as required.

[p. 158] It follows that there is no separation $(B_1, B_2)$ of $B$ of order 0 with $B_1, B_2$ non-null. Suppose that $e \in E(B)$ has all its ends in $V(A)$. Let $K_e$ be the hypergraph with edge set $\{e\}$ and vertex set the set of ends of $e$; then $(K_e, B \setminus e)$ is a separation of $B$. Now $K_e \not\subseteq A$ since $e \notin E(A)$, and $B \setminus e \not\subseteq A$

since $V(A) \neq V(G)$, and so

$$|V(K_e \cap (B \setminus e))| > \min(|V(A \cap K_e)|, |V(A \cap (B \setminus e))|).$$

But the left side is the number of ends of $e$, and so is the right side, a contradiction. Thus there is no such $e$. □

(2.9) *Let $\mathscr{T}$ be a tangle of order $\theta$ in a hypergraph $G$, and let $(A_1, B_1) \in \mathscr{T}$. Let $(A_2, B_2)$ be a separation of order $< \theta$. If either*

(i) $V(B_1) \subseteq V(B_2)$, or

(ii) $V(A_2) \subseteq V(A_1)$, or

(iii) $\theta \geq 2$ and $E(A_2) \subseteq E(A_1)$ (equivalently, $E(B_1) \subseteq E(B_2)$)

*then $(A_2, B_2) \in \mathscr{T}$.*

*Proof.* Suppose not; then $(B_2, A_2) \in \mathscr{T}$. Choose $(A, B) \in \mathscr{T}$, extreme, with $B_2 \subseteq A$ and $B \subseteq A_2$. Then $A \cup A_1 \neq G$ by the second axiom. Since $A \cup B = G$ and $A_1 \cup B_1 = G$ it follows that $B \not\subseteq A_1$ and $B_1 \not\subseteq A$.

*Case* 1. $V(B_1) \subseteq V(B_2)$.

Then $V(B_1) \subseteq V(B_2) \subseteq V(A)$, and $E(B_1) \cap E(B) = \emptyset$, since from (2.8) every edge of $B$ has an end in $V(G) - V(A) \subseteq V(G) - V(B_1)$. Thus $E(B_1) \subseteq E(A)$ and so $B_1 \subseteq A$, a contradiction.

*Case* 2. $V(A_2) \subseteq V(A_1)$.

Since $(B_2, A_2) \in \mathscr{T}$ and $(B_1, A_1)$ has order $< \theta$, and $V(A_2) \subseteq V(A_1)$, it follows that $(B_1, A_1) \in \mathscr{T}$, since the theorem holds in Case 1. But this contradicts (2.1).

*Case* 3. $\theta \geq 2$ and $E(A_2) \subseteq E(A_1)$.

Since $E(B) \subseteq E(A_2) \subseteq E(A_1)$ and $B \not\subseteq A_1$, there is a vertex $v$ of $B$ with $v \notin V(A_1)$. Since $E(B) \subseteq E(A_1)$, it follows that $v$ is incident with no edge of $B$. By (2.8), $V(B) = \{v\}$ and $E(B) = \emptyset$, and since $V(A) \neq V(G)$, it follows that $V(A \cap B) = \emptyset$. By (2.8) again, $\theta = 1$, a contradiction. □

For future reference, we observe the following.

(2.10) *Let $\mathscr{T}$ be a tangle of order $\geq 3$ in a graph $G$, and let $(A, B) \in \mathscr{T}$. Then $B$ has a circuit.*

[p. 159] *Proof.* It suffices to prove the result when $(A, B)$ is extreme. By (2.8), $|A \cap B| \geq 2$; let $v_1, v_2 \in V(A \cap B)$ be distinct.

(1) *There is no separation $(B_1, B_2)$ of $B$ of order $\leq 1$ with $v_1 \in V(B_1) - V(B_2)$ and $v_2 \in V(B_2) - V(B_1)$.*

For such a separation would satisfy

$$\min(|V(A \cap B_1)|, |V(A \cap B_2)|) \geq 1$$

and $B_1, B_2 \neq B$, contrary to (2.8).

Moreover, from (2.8), $v_1$ and $v_2$ are not adjacent in $B$. From (1) and Menger's theorem, there are two paths of $B$ between $v_1$ and $v_2$, internally disjoint, and hence $B$ has a circuit, as required. □

## 3. A Lemma About Submodular Functions

Now we turn to our first main result, the minimax theorem relating tangle number and branch-width. It is most convenient to prove a generalization, which is a statement about submodular functions.

Let $E$ be a finite set. A *connectivity function* on $E$ is a function $\kappa$ from the set of all subsets of $E$ to the set of integers such that (i) for $X \subseteq E$, $\kappa(X) = \kappa(E - X)$ (ii) for $X, Y \subseteq E$, $\kappa(X \cup Y) + \kappa(X \cap Y) \leq \kappa(X) + \kappa(Y)$.

For instance, if $G$ is a hypergraph and $E = E(G)$, we would let $\kappa(X)$ be the number of vertices of $G$ incident both with an edge in $X$ and with an edge in $E - X$; or if $M$ is a matroid with rank function $r$ and $E = E(M)$, we could let $\kappa(X) = r(X) + r(E - X)$.

A subset $X \subseteq E$ is *efficient* if $\kappa(X) \leq 0$. A *bias* is a set $\mathscr{B}$ of efficient sets, such that (i) if $X \subseteq E$ is efficient then $\mathscr{B}$ contains one of $X, E - X$ (ii) if $X, Y, Z \in \mathscr{B}$ then $X \cup Y \cup Z \neq E$.

A bias $\mathscr{B}$ is said to *extend* a set $\mathscr{A}$ of efficient sets if $\mathscr{A} \subseteq \mathscr{B}$. We are concerned with the problem of, given $\mathscr{A}$, when is there a bias extending $\mathscr{A}$?

Let us describe an obstacle to the existence of such a bias. A tree is a connected non-null graph with no circuits; its vertices of valency $\leq 1$ are its *leaves*. A tree is *ternary* if every vertex has valency 1 or 3. (Thus, ternary trees have $\geq 2$ leaves.) An *incidence* in a tree $T$ is a pair $(v, e)$, where $v \in V(T)$, $e \in E(T)$, and $e$ is incident with $v$. A *tree-labelling over* $\mathscr{A}$ is a pair $(T, \alpha)$, where $T$ is a ternary tree, and $\alpha$ is a function from the set of all incidences in $T$ to the set of efficient subsets of $E$, such that

[p. 160] (i) for each $e \in E(T)$ with ends $u, v$, say, $\alpha(u, e) = E - \alpha(v, e)$

(ii) for each incidence $(v, e)$ in $T$ such that $v$ is a leaf, either $\alpha(v, e) = E$ or $\alpha(v, e) \cup X = E$ for some $X \in \mathscr{A}$

(iii) if $v \in V(T)$ has valency 3, incident with $e_1, e_2, e_3$, say, then $\alpha(v, e_1) \cup \alpha(v, e_2) \cup \alpha(v, e_3) = E$.

(3.1) *If there is a bias extending $\mathscr{A}$ then there is no tree-labelling over $\mathscr{A}$.*

*Proof.* Suppose that $\mathscr{B}$ is a bias extending $\mathscr{A}$, and $(T, \alpha)$ is a tree-labelling over $\mathscr{A}$. An incidence $(v, e)$ of $T$ is *passive* if $\alpha(v, e) \notin \mathscr{B}$. For each edge $e$ with ends $u, v$, $\mathscr{B}$ contains exactly one of $\alpha(u, e)$, $\alpha(v, e)$ since they are efficient complementary sets. Thus there are precisely $|E(T)|$ passive incidences. Since $T$ has $|E(T)| + 1$ vertices there is a vertex $v$ of $T$ in no

passive incidence; that is, $\alpha(v,e) \in \mathscr{B}$ for all edges $e$ incident with $v$. If $v$ has valency 1 then by the definition of a tree-labelling, either $\alpha(v,e) = E$ or $\alpha(v,e) \cup X = E$ for some $X \in \mathscr{A}$, in either case contrary to the definition of a bias. Thus $v$ has valency 3. Let $e_1, e_2, e_3$ be the edges of $T$ incident with $v$; then

$$\alpha(v,e_1) \cup \alpha(v,e_2) \cup \alpha(v,e_3) = E$$

by the definition of a tree-labelling, and yet each $\alpha(v,e_i) \in \mathscr{B}$, contrary to the definition of a bias, as required. □

The main result of this section is a converse of (3.1), in a strong form, that if there is no bias extending $\mathscr{A}$, then there is an exact tree-labelling over $\mathscr{A}$. "Exact" is defined as follows. Let $(T,\alpha)$ be a tree-labelling over $\mathscr{A}$. A *fork* in $T$ is an unordered pair $\{e_1,e_2\}$ of distinct edges of $T$ with a common end (the *nub* of the fork). A fork $\{e_1,e_2\}$ with nub $t$ is *exact* (for $\alpha$) if $\alpha(t,e_1) \cap \alpha(t,e_2) = \emptyset$. We say that $(T,\alpha)$ is *exact* if every fork of $T$ is exact. We require the following lemma.

(3.2) *If there is a tree-labelling over $\mathscr{A}$ then there is an exact tree-labelling over $\mathscr{A}$, using the same tree.*

*Proof.* Choose a tree $T$ such that there is a tree-labelling $(T,\alpha)$ over $\mathscr{A}$. Choose $t_0 \in V(T)$. For each $t \in V(T)$ we denote by $d(t)$ the number of edges in the path of $T$ between $t_0$ and $t$. Choose $\alpha$ satisfying (1), (2), and (3), below.

(1) *$(T,\alpha)$ is a tree-labelling over $\mathscr{A}$.*

(2) *Subject to (1), $\sum \kappa(\alpha(v,e))$ (summed over all incidences $(v,e)$ of $T$) is minimum.*

(3) *Subject to (1) and (2), $\sum 3^{-d(t)}$ (summed over all non-exact forks, where $t$ is the nub of the fork) is minimum.*

[p. 161] We claim that $(T,\alpha)$ is exact. For suppose that some fork $\{e_1,e_2\}$ with nub $t$ is non-exact. Then $t$ has valency 3 in $T$, since $T$ is ternary; let $e_3$ be the third edge of $T$ incident with $v$, and let $e_i$ have ends $t, t_i$ $(i = 1,2,3)$. Let $A_1 = \alpha(t,e_1)$, $A_2 = \alpha(t,e_2)$. Define $\alpha'$ by

$$\begin{aligned} \alpha'(t,e_1) &= A_1 - A_2 \\ \alpha'(t_1,e_1) &= \alpha(t_1,e_1) \cup A_2 = E - (A_1 - A_2) \\ \alpha'(v,e) &= \alpha(v,e) \text{ for } (v,e) \neq (t,e_1),(t_1,e_1). \end{aligned}$$

We claim that $\kappa(A_1 - A_2) \geq \kappa(A_1)$. For if $\kappa(A_1 - A_2) \geq 0$ this is true, and so we may assume that $A_1 - A_2$ is efficient. Then $\alpha'$ is a tree-labelling over $\mathscr{A}$, and from (2),

$$\kappa(\alpha'(t,e_1)) + \kappa(\alpha'(t_1,e_1)) \geq \kappa(\alpha(t,e_1)) + \kappa(\alpha(t_1,e_1));$$

that is,

$$\kappa(A_1 - A_2) + \kappa(E - (A_1 - A_2)) \geq \kappa(A_1) + \kappa(E - A_1).$$

Since $\kappa(E - (A_1 - A_2)) = \kappa(A_1 - A_2)$ and $\kappa(E - A_1) = \kappa(A_1)$, it follows that $\kappa(A_1 - A_2) \geq \kappa(A_1)$, as claimed. Similarly $\kappa(A_2 - A_1) \geq \kappa(A_2)$. But since $\kappa$ is a connectivity function,

$$\kappa(A_1) + \kappa(E - A_2) \geq \kappa(A_1 \cup (E - A_2)) + \kappa(A_1 \cap (E - A_2));$$

that is,

$$\kappa(A_1) + \kappa(A_2) \geq \kappa(A_2 - A_1) + \kappa(A_1 - A_2).$$

Thus equality holds throughout, and in particular, $\kappa(A_1 - A_2) = \kappa(A_1)$ and $\kappa(A_2 - A_1) = \kappa(A_2)$. From the symmetry between $t_1$ and $t_2$, we may assume that $d(t) < d(t_1)$. With $\alpha'$ as before we see that $\alpha'$ is a tree-labelling over $\mathscr{A}$ and $\sum \kappa(\alpha'(v, e)) = \sum \kappa(\alpha(v, e))$. Moreover, $\{e_1, e_2\}$ is exact for $\alpha'$, and any fork of $T$ which is exact for $\alpha$ is exact for $\alpha'$ except possibly for forks $\{e, e_1\}$ with nub $t_1$. There are at most two such forks, and since $d(t_1) > d(t)$, this contradicts (3), as required. □

(3.3) *Let $(T, \alpha)$ be an exact tree-labelling over $\mathscr{A}$, and let $(u, f)$ be an incidence in $T$. Let $T_0$ be the component of $T \setminus f$ which contains $u$. Then, as $(v, e)$ ranges over all incidences of $T$ such that $v$ is a leaf of $T$ and $v \in V(T_0)$, the sets $E - \alpha(v, e)$ are mutually disjoint and have union $E - \alpha(u, f)$.*

*Proof.* We proceed by induction on $|V(T_0)|$. If $u$ is a leaf the result is trivial, and so we may assume that $u$ is incident with three edges $f, f_1, f_2$; let $f_i$ have ends $u$, $u_i$ $(i = 1, 2)$, and let $T_i$ be the component of [p. 162] $T \setminus f_i$ containing $u_i$ $(i = 1, 2)$. Then $V(T_0) = V(T_1) \cup V(T_2) \cup \{u\}$ and $V(T_1) \cap V(T_2) = \emptyset$. Now the result holds for $(u_1, f_1)$ and $(u_2, f_2)$ by the inductive hypothesis. Moreover, since $E - \alpha(u_i, f_i) = \alpha(u, f_i)$ $(i = 1, 2)$ and $(T, \alpha)$ is exact, it follows that

$$(E - \alpha(u_1, f_1)) \cup (E - \alpha(u_2, f_2)) = E - \alpha(u, f)$$
$$(E - \alpha(u_1, f_1)) \cap (E - \alpha(u_2, f_2)) = \emptyset.$$

The result follows. □

(3.4) *If there is no bias extending $\mathscr{A}$ then there is an exact tree-labelling over $\mathscr{A}$.*

*Proof.* By (3.2), it suffices to prove that there is a tree-labelling over $\mathscr{A}$. Suppose that $E = \emptyset$. If $\emptyset$ is efficient, let $T$ be a two-vertex tree, and

let $\alpha(v,e) = \emptyset$ for both incidences $(v,e)$ of $T$; $(T,\alpha)$ is the required tree-labelling. If $\emptyset$ is not efficient, then $\mathscr{A}$ is a bias, a contradiction. Thus we may assume that $E \neq \emptyset$. Choose $x \in E$, and let $\mathscr{B}$ be the set of all efficient sets $B \subseteq E$ with $x \notin B$; then $\mathscr{B}$ is a bias. Since $\mathscr{B}$ does not extend $\mathscr{A}$, it follows that $\mathscr{A} \neq \emptyset$.

We proceed by induction on the number $N$ of efficient sets $X \subseteq E$ such that neither $X$ nor $E - X$ is a subset of any member of $\mathscr{A}$. We suppose first that $N = 0$. Let $\mathscr{B}$ be the set of all efficient sets which are subsets of members of $\mathscr{A}$. Since $\mathscr{A} \subseteq \mathscr{B}$, $\mathscr{B}$ is not a bias. But for every efficient set $X$, either $X \in \mathscr{B}$ or $E - X \in \mathscr{B}$ since $N = 0$. Thus there exist $X_1, X_2, X_3 \in \mathscr{B}$ with $X_1 \cup X_2 \cup X_3 = E$. Let $T$ be the tree with four vertices $t_0$, $t_1$, $t_2$, $t_3$ and edges $e_i$ with ends $t_0$, $t_i$ $(i = 1,2,3)$. Define $\alpha(t_0, e_i) = X_i$, $\alpha(t_i, e_i) = E - X_i$ $(i = 1,2,3)$. Then $(T,\alpha)$ is a tree-labelling over $\mathscr{A}$, as required.

Thus we may assume $N > 0$. Choose an efficient set $X \subseteq E$ such that neither $X$ nor $E - X$ is a subset of any member of $\mathscr{A}$, and subject to that with $X$ minimal. Since $\mathscr{A} \neq \emptyset$, $X \neq \emptyset$. Let $\mathscr{A}_1 = \mathscr{A} \cup \{X\}$, $\mathscr{A}_2 = \mathscr{A} \cup \{E - X\}$. Since there is no bias extending $\mathscr{A}$, there is no bias extending $\mathscr{A}_1$ or $\mathscr{A}_2$. From our inductive hypothesis there are exact tree-labellings $(T_1, \alpha_1)$ over $\mathscr{A}_1$ and $(T_2, \alpha_2)$ over $\mathscr{A}_2$. A leaf $t$ of $T_1$ is bad if $\alpha_1(t,e) \neq E$ and $\alpha_1(t,e) \cup A \neq E$ for all $A \in \mathscr{A}$, where $(t,e)$ is an incidence, and we define the bad leaves of $T_2$ similarly. Now if $t$ is a bad leaf of $T_1$ and $(t,e)$ is an incidence, then $\alpha_1(t,e) \cup X = E$ and so $E - \alpha_1(t,e) \subseteq X$. If $E - \alpha_1(t,e) \neq X$, then from our choice of $X$, either $E - \alpha_1(t,e) \subseteq A$ for some $A \in \mathscr{A}$ or $\alpha_1(t,e) \subseteq A$ for some $A \in \mathscr{A}$. In the first case $\alpha_1(t,e) \cup A = E$, a contradiction, since $t$ is bad. In the second case $E - X \subseteq \alpha_1(t,e) \subseteq A$, contrary to our choice of $X$. Thus $E - \alpha_1(t,e) = X$, for every bad leaf $t$. Since $X \neq \emptyset$, it follows from (3.3) that there is at most one bad leaf in $T_1$.

[p. 163] On the other hand, we may assume that $T_1$ has at least one bad leaf, for otherwise $(T_1, \alpha_1)$ is the desired tree-labelling over $\mathscr{A}$. Let $t_0$ be the unique bad leaf of $T_1$, incident with an edge $e_0$. Then $\alpha_1(t_0, e_0) = E - X$. Let the ends of $e_0$ be $t_0, s$. Then $\alpha_1(s, e_0) = X$. Since $X \neq E$ and $E - X$ is not a subset of any member of $\mathscr{A}_1$, $s$ is not a leaf of $T_1$. Let $S = T_1 \setminus t_0$; then $s$ has valency 2 in $S$.

Let the bad leaves of $T_2$ be $t_1, \ldots, t_r$, incident with edges $e_1, \ldots, e_r$, respectively. Then as before

$$\alpha_2(t_i, e_i) \cup (E - X) = E,$$

that is, $X \subseteq \alpha_2(t_i, e_i)$, for $1 \leq i \leq r$. Let $S^1, \ldots, S^r$ be $r$ copies of $S$, mutually disjoint. For $v \in V(S)$ and $e \in E(S)$ let $v^i$ and $e^i$ denote the

corresponding vertex and edge of $S^i$ $(1 \le i \le r)$. Choose $S^1, \dots, S^r$ so that $s^i = t_i$ and $V(S^i) \cap V(T_2) = t_i$ $(1 \le i \le r)$, and let $T$ be the tree formed by the union of $T_2$ and $S^1, \dots, S^r$. Every incidence of $T$ is an incidence of exactly one of $T_2$, $S^1, \dots, S^r$. We define $\alpha$ by

$$\begin{aligned} \alpha(v,e) &= \alpha_2(v,e) && \text{if } (v,e) \text{ is an incidence of } T_2 \\ \alpha(v^i,e^i) &= \alpha_1(v,e)\ (1 \le i \le r) && \text{if } (v,e) \text{ is an incidence of } T_1. \end{aligned}$$

We claim that $(T,\alpha)$ is a tree-labelling over $\mathscr{A}$, and this follows easily from the fact that

$$\alpha_1(s,e_0) = X \subseteq \alpha_2(t_i,e_i) \quad (1 \le i \le r).$$

Then the result follows. □

In summary then we have shown

(3.5) *The following are equivalent:* (i) *there is no bias extending* $\mathscr{A}$ (ii) *there is a tree-labelling over* $\mathscr{A}$ (iii) *there is an exact tree-labelling over* $\mathscr{A}$.

We observe also

(3.6) *If there is an exact tree-labelling over* $\mathscr{A}$*, then either* $E = \emptyset$*, or* $E \in \mathscr{A}$*, or there is an exact tree-labelling* $(T,\alpha)$ *over* $\mathscr{A}$ *such that for each leaf* $v$ *and incident edge* $e$*,* $\alpha(v,e) \ne E$.

*Proof.* Choose an exact tree-labelling $(T,\alpha)$ with $|V(T)|$ minimum. Suppose that for some leaf $v_0$ and incident edge $e_0$, $\alpha(v_0,e_0) = E$. Let $v$ be [p. 164] the other end of $e_0$. Then $\alpha(v,e_0) = \emptyset$. If $v$ is also a leaf, then either $E = \emptyset$ or $E \in \mathscr{A}$, as required. We assume then that $v$ has two other neighbours $v_1, v_2$ in $T$; let $e_i$ be the edge joining $v$ and $v_i$ $(i = 1,2)$. Now since $(T,\alpha)$ is exact, $\alpha(v,e_0)$, $\alpha(v,e_1)$, $\alpha(v,e_2)$ are mutually disjoint and have union $E$. Since $\alpha(v,e_0) = \emptyset$, it follows that $\alpha(v_1,e_1) = \alpha(v,e_2)$ and $\alpha(v,e_1) = \alpha(v_2,e_2)$. Let $T'$ be obtained from $T$ by deleting $v$ and $v_0$ and adding a new edge $f$ joining $v_1$ and $v_2$. We define $\alpha'(v_1,f) = \alpha(v_1,e_1)$, $\alpha'(v_2,f) = \alpha(v_2,e_2)$, and otherwise $\alpha' = \alpha$; then $(T',\alpha')$ is an exact tree-labelling over $\mathscr{A}$ with $|V(T')| < |V(T)|$, a contradiction. □

## 4. Branch-width

A *branch-decomposition* of a hypergraph $G$ is a pair $(T,\tau)$, where $T$ is a ternary tree and $\tau$ is a bijection from the set of leaves of $T$ to $E(G)$. The *order* of an edge $e$ of $T$ is the number of vertices $v$ of $G$ such that there are leaves $t_1, t_2$ of $T$ in different components of $T \setminus e$, with $\tau(t_1), \tau(t_2)$ both incident with $v$. The *width* of $(T,\tau)$ is the maximum order of the edges of $T$, and the *branch-width* $\beta(G)$ of $G$ is the minimum width of all

branch-decompositions of $G$ (or 0 if $|E(G)| \leq 1$, when $G$ has no branch-decompositions). For example, Fig. 1 shows a branch-decomposition with width 2 of a series-parallel graph.

Let us prove some properties of branch-width. A graph $H$ is a *minor* of a graph $G$ if $H$ can be obtained from a subgraph of $G$ by contracting edges.

(4.1) *If $H$ is a minor of a graph $G$, then $\beta(H) \leq \beta(G)$.*

*Proof.* We may assume that $|E(H)| \geq 2$, for otherwise $\beta(H) = 0$. Let $(T, \tau)$ be a branch-decomposition of $G$ with width $\beta(G)$. Let $S$ be a minimal subtree of $T$ such that $\tau^{-1}(e) \in V(S)$ for all $e \in E(H)$, and let $T'$ be obtained from $S$ by suppressing all vertices of valency 2 (that is, for any vertex of valency 2 we delete it and add an edge joining its neighbours and continue this process until no such vertices remain). Let $\tau'$ be the restriction of $\tau$ to the set of leaves of $T'$; then $(T', \tau')$ is a branch-decomposition of $H$, and its width is $\leq \beta(G)$, as is easily seen. The result follows. □

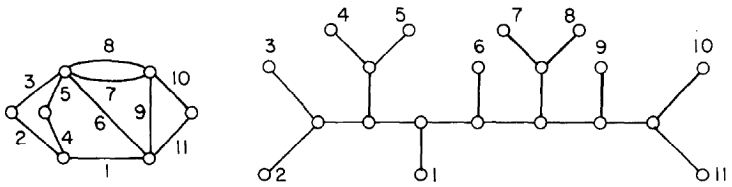



FIGURE 1

[p. 165] (4.2) *A graph $G$ has branch-width*

(i) *0 if and only if every component of $G$ has $\leq 1$ edge*

(ii) *$\leq 1$ if and only if every component of $G$ has $\leq 1$ vertex of valency $\geq 2$*

(iii) *$\leq 2$ if and only if $G$ has no $K_4$ minor.*

*Proof.* Statement (i) is clear. The "if" part of (ii) is easy and "only if" follows from (4.1) and the fact that a 2-edge circuit and a 3-edge path both have branch-width 2, when $G$ has no loops. In general, let a component of $G$ have two vertices of valency $\geq 2$. The first edge $e$ of a path between them joins two vertices of valency $\geq 2$. So in every branch-decomposition $(T, \tau)$ of $G$, the edge of $T$ incident with the leaf $\tau^{-1}(e)$ has order 2, and $G$ has branch-width $\geq 2$.[F2] The "only if" part of (iii) follows similarly, while

[F2] **Proof gap.** The printed argument misses graphs with loops. Two vertices joined by one edge, with a loop at each, have two vertices of valency $\geq 2$, but no 2-edge circuit or 3-edge path as a minor, since these need two edges that are not loops. The inserted argument proves "only if" directly and covers loops. The statement of (4.2) holds as printed.

the "if" part may be proved by induction on the size of $G$, using Dirac's theorem [1] that any non-null simple graph with no $K_4$ minor has a vertex of valency $\leq 2$. $\square$

The main result of this section is the following. We denote by $\gamma(G)$ the maximum size of an edge of $G$ (setting $\gamma(G) = 0$ if $E(G) = \emptyset$). We recall that $\theta(G)$ is the tangle number of $G$.

(4.3) *For any hypergraph $G$, $\max(\beta(G), \gamma(G)) = \theta(G)$ unless $\gamma(G) = 0$ and $V(G) \neq \emptyset$.*

*Proof.* Suppose first that $\gamma(G) = 0$ and that $\mathscr{T}$ is a tangle in $G$ of order $\geq 2$. Choose $(A, B) \in \mathscr{T}$, extreme. By (2.8), $E(B) = \emptyset$, and so $E(A) = E(G)$, contrary to (2.3). Thus, if $\gamma(G) = 0$ then $\theta(G) \leq 1$. Moreover, if $\gamma(G) = 0$ then $\beta(G) = 0$, and $\theta(G) = 1$ if and only if $V(G) \neq \emptyset$, by (2.5). Thus if $\gamma(G) = 0$ the result holds, and we henceforth assume that $\gamma(G) > 0$.

Let $E = E(G)$, and for $X \subseteq E$, define $\kappa_0(X)$ to be the number of vertices of $G$ incident both with an edge in $X$ and with an edge in $E - X$. Choose $k \geq \gamma(G)$, and let $\kappa(X) = \kappa_0(X) - k$. It is easily seen that $\kappa$ is a connectivity function, and for every $e \in E(G)$, $\{e\}$ is efficient. Let $\mathscr{A} = \{\{e\} : e \in E(G)\}$.

(1) *There is a bias extending $\mathscr{A}$ if and only if $G$ has a tangle of order $k + 1$.*

For if $\mathscr{T}$ is a tangle in $G$ of order $k + 1$, let $\mathscr{B} = \{E(A) : (A, B) \in \mathscr{T}\}$. Then $B$ is a bias, by (2.3), since $k \geq \gamma(G) \geq 1$, and it extends $\mathscr{A}$ by the third axiom. For the converse, let $\mathscr{B}$ be a bias extending $\mathscr{A}$, and let $\mathscr{T}$ be the set of all separations $(A, B)$ of $G$ of order $\leq k$ with $E(A) \in \mathscr{B}$. We claim that $\mathscr{T}$ is a tangle of order $k + 1$. For if $(A, B)$ is a separation of $G$ of order $\leq k$, then $E(A)$ and $E(B)$ are both efficient, and so one of $E(A)$, $E(B)$ is in $\mathscr{B}$, $E(A)$, say; but then $(A, B) \in \mathscr{T}$. Thus the first axiom holds, and clearly so does the second. Since $k \geq \gamma(G)$ and $\mathscr{B}$ extends $\mathscr{A}$, $(K_e, G \setminus e) \in \mathscr{T}$ for every $e \in E$, where $K_e$ is the hypergraph consisting of $e$ and its ends. By (2.7), $\mathscr{T}$ is a tangle of order $k + 1$, as required.

(2) *There is an exact tree-labelling over $\mathscr{A}$ if and only if $\beta(G) \leq k$.*

[p. 166] For if $|E| \leq 1$, then $\beta(G) = 0 \leq k$ and there is an exact tree-labelling over $\mathscr{A}$, and so we may assume that $|E| \geq 2$. If $(T, \tau)$ is a branch-decomposition of $G$ of width $\leq k$, define $\alpha(v, e)$ for each incidence $(v, e)$ to be the set of all edges $\tau(t)$ of $G$ with $t$ and $v$ in different components of $T \setminus e$. Then $(T, \alpha)$ is an exact tree-labelling over $\mathscr{A}$. For the converse, suppose that there is an exact tree-labelling over $\mathscr{A}$. Since $|E| > 1$, it follows that $E \notin \mathscr{A}$ and $E \neq \emptyset$, and so by (3.6) we may choose an exact tree-labelling $(T, \alpha)$ over $\mathscr{A}$ such that for each leaf $v$ and incident edge $e$, $\alpha(v, e) \neq E$. For such $v$, $e$, there exists $\{f\} \in \mathscr{A}$ such that $\alpha(v, e) = E - \{f\}$; we define $f = \tau(v)$.

By (3.3), $(T, \tau)$ is a branch-decomposition of $G$ of width $\leq k$.

From (3.5), (1), and (2) we deduce that

(3) *For all $k \geq \gamma(G)$, $G$ has a tangle of order $k+1$ if and only if $k < \beta(G)$.*

Now we deduce the theorem. By (2.4), $\theta(G) \geq \gamma(G)$. By setting $k = \theta(G)$ we deduce from (3) that $\beta(G) \leq \theta(G)$, and so $\max(\beta(G), \gamma(G)) \leq \theta(G)$. By setting $k = \theta(G) - 1$ we deduce from (3) that $\theta(G) \leq \max(\beta(G), \gamma(G))$. The result follows. □

We apply (4.3) (actually, the easy part of (4.3)) for the following.

(4.4) *For $n \geq 0$, $K_n$ has tangle number $\lceil (2/3)n \rceil$, and for $n \geq 3$, it has branch-width $\lceil (2/3)n \rceil$.*

*Proof.* The result holds for $n \leq 3$, and we assume that $n \geq 4$. Put $\theta = \lceil (2/3)n \rceil$. It is easy to see that $K_n$ has a branch-decomposition of width $\leq \theta$. Thus the result follows from (4.3) if we can find a tangle of order $\theta$. Let $\mathscr{T}$ be the set of all separations $(A, B)$ of $G = K_n$ with $|V(A)| < \theta$. If $(A, B)$ is any separation of $G$ then one of $V(A)$, $V(B)$ equals $V(G)$, and so its order equals the smaller of $|V(A)|$, $|V(B)|$. Hence if $(A, B)$ has order $< \theta$ then $\mathscr{T}$ contains one of $(A, B)$, $(B, A)$, and the first axiom is satisfied. For the second axiom, suppose that $(A_i, B_i) \in \mathscr{T}$ $(1 \leq i \leq 3)$ and $A_1 \cup A_2 \cup A_3 = G$. Since

$$|V(A_1)| + |V(A_2)| + |V(A_3)| \leq 3\theta - 3 < 2n$$

some vertex $v$ of $G$ is in at most one of $V(A_1)$, $V(A_2)$, $V(A_3)$; $v \notin V(A_1) \cup V(A_2)$, say. Since $|V(A_3)| < \theta < n$ some vertex $u$ of $G$ is not in $V(A_3)$. But then the edge joining $u$ and $v$ is in none of $E(A_1)$, $E(A_2)$, $E(A_3)$, a contradiction. Thus the second axiom is satisfied. For the third, let $e \in E(G)$, and let $K$ be the graph formed by $e$ and its ends; then $(K, G \setminus e) \in \mathscr{T}$ by definition of $\mathscr{T}$, since $\theta \geq 3$, and so $\mathscr{T}$ is a tangle by (2.7). This completes the proof. □

[p. 167] Let us mention the following weakening of the second tangle axiom.

(4.5) *Let $\theta \geq 2$, and let $\mathscr{T}$ be a set of separations of a hypergraph $G$, each of order $< \theta$. Suppose that the first tangle axiom holds, and*

(i) *if $(A_1, B_1), (A_2, B_2) \in \mathscr{T}$ then $B_1 \not\subseteq A_2$*

(ii) *there do not exist subhypergraphs $A_1, A_2, A_3 \subseteq G$, mutually edge-disjoint, with $A_1 \cup A_2 \cup A_3 = G$ and with $(A_1, A_2 \cup A_3)$, $(A_2, A_3 \cup A_1)$, $(A_3, A_1 \cup A_2)$ all in $\mathscr{T}$.*

*Then the second tangle axiom holds.*

*Proof.* Suppose that the second axiom fails, and choose $(A_1, B_1), (A_2, B_2), (A_3, B_3) \in \mathscr{T}$ such that $A_1 \cup A_2 \cup A_3 = G$, satisfying

(1) $\sum_{1\le i\le 3}|V(A_i\cap B_i)|$ is minimum, and

(2) subject to (1), $A_1, A_2, A_3$ are minimal.

We observe

(3) *For $1\le i\le 3$, if $v\in V(A_i\cap B_i)$ then $v$ is incident with an edge of $B_i$; and also with an edge of $A_i$, unless $v$ belongs to no other $A_j$ $(j\ne i)$.*

For if $v$ is incident with no edge of $B_i$ then $(A_i, B_i\setminus v)$ is a separation, and it belongs to $\mathscr{T}_1$ by the first axiom and (i), contrary to (1). If $v$ is incident with no edge of $A_i$ then $(A_i\setminus v, B_i)$ is a separation, and it belongs to $\mathscr{T}$, by the first axiom and (i), and so by (1), $v$ belongs to no $A_j$ $(j\ne i)$.

(4) *For $1\le i,j\le 3$ with $i\ne j$, $A_i\subseteq B_j$.*

For let $i=1$, $j=2$, say. The sum of the orders of $(A_1\cap B_2, B_1\cup A_2)$ and $(A_1\cup B_2, B_1\cap A_2)$ equals the sum of the orders of $(A_1,B_1)$ and $(A_2,B_2)$. If $(A_1\cap B_2, B_1\cup A_2)$ has order at most that of $(A_1,B_1)$, then since $(A_1\cap B_2)\cup A_2\cup A_3 = G$ and $(A_1\cap B_2, B_1\cup A_2)\in\mathscr{T}$ by the first axiom and (i), it follows from (2) that $A_1\cap B_2 = A_1$; that is, $A_1\subseteq B_2$. Thus $E(A_2)\subseteq E(B_1)$. Suppose that $A_2\not\subseteq B_1$, and choose $v\in V(A_2)-V(B_1)$. Then $v\in V(A_1\cap A_2)$, and by (3), $v$ is incident with an edge in $E(A_2)\subseteq E(B_1)$; yet $v\notin V(B_1)$, a contradiction. Thus $A_2\subseteq B_1$. Similarly, if $(A_2\cap B_1, B_2\cup A_1)$ has order at most that of $(A_2,B_2)$, then $A_1\subseteq B_2$ and $A_2\subseteq B_1$. The result follows, since one of these inequalities must apply.

From (4), $A_1\cup A_2\subseteq B_3$, and so $(A_3, A_1\cup A_2)$ is a separation of order $<\theta$. Since $(A_3,B_3)\in\mathscr{T}$, it follows from (i) that $(A_1\cup A_2, A_3)\notin\mathscr{T}$, and so $(A_3, A_1\cup A_2)\in\mathscr{T}$, from the first axiom. Similarly $(A_1, A_2\cup A_3)$, $(A_2, A_3\cup A_1)\in\mathscr{T}$, contrary to (ii). □

## [p. 168] 5. Branch-width And Tree-width

A tree-decomposition of a hypergraph $G$ is a pair $(T,\tau)$, where $T$ is a tree and for $t\in V(T)$, $\tau(t)$ is a subhypergraph of $G$ with the following properties:

(i) $\bigcup(\tau(t) : t\in V(T)) = G$

(ii) for distinct $t,t'\in V(T)$, $E(\tau(t)\cap\tau(t')) = \emptyset$

(iii) for $t,t',t''\in V(T)$, if $t'$ is on the path of $T$ between $t$ and $t''$ then $\tau(t)\cap\tau(t'')\subseteq\tau(t')$.

The width of such a tree-decomposition is the maximum of $(|V(\tau(t))|-1)$, taken over all $t\in V(T)$, and the tree-width $\omega(G)$ of $G$ is the minimum width of all tree-decomposition of $G$. (Thus, $\omega(G)\ge 0$ unless $V(G)=\emptyset$, when $\omega(G)=-1$.)

Let us compare tree-width and branch-width.

(5.1) *For any hypergraph $G$,* $\max(\beta(G),\gamma(G))\le\omega(G)+1\le\max(\lfloor(3/2)\beta(G)\rfloor,\gamma(G),1)$.

*Proof.* If $\gamma(G) = 0$ then $\beta(G) = 0$ and $\omega(G) \leq 0$, and the result holds. We assume then that $\gamma(G) > 0$, and so $V(G) \neq \emptyset$ and $E(G) \neq \emptyset$. If $|E(G)| = 1$ then $\beta(G) = 0$ and $\omega(G) = \gamma(G) - 1$, and again the result holds. Thus we may assume that $|E(G)| \geq 2$. Since the removal of isolated vertices does not change any of $\beta$, $\gamma$, $\omega$, we may assume that there are no isolated vertices in $G$. We show the second inequality first.

Let $(T, \tau)$ be a branch-decomposition of $G$ of width $\beta(G)$. For each $t \in V(T)$ we define a subhypergraph $\sigma(t)$ of $G$ as follows: (i) if $t$ is a leaf of $T$, let $\sigma(t)$ be the hypergraph consisting of $\tau(t)$ and its ends (ii) if $t$ is not a leaf of $T$, let $U_t$ consist of those vertices $v$ of $G$ for which there are edges $f, g$ of $G$, both incident with $v$, such that $t$ lies on the path of $T$ between $\tau^{-1}(f)$ and $\tau^{-1}(g)$. Let $V(\sigma(t)) = U_t$, $E(\sigma(t)) = \emptyset$.

It is easy to verify that $(T, \sigma)$ is a tree-decomposition of $G$. Let us bound its width. If $t$ is a leaf of $T$, $|V(\sigma(t))| \leq \gamma(G)$. If $t$ is not a leaf of $T$, let $e_1, e_2, e_3$ be the three edges of $T$ incident with $t$. For any $v \in U_t$, $v$ contributes to the order of at least two of $e_1, e_2, e_3$, and so $2|U_t| \leq 3\beta(G)$. Thus, this tree-decomposition has width $\leq \max(\gamma(G), (3/2)\beta(G)) - 1$, and so $\omega(G) + 1 \leq \max(\gamma(G), (3/2)\beta(G))$, as required.

Now we show the first inequality. Clearly $\gamma(G) \leq \omega(G) + 1$. Let $(T, \tau)$ be a tree-decomposition of $G$ of width $\omega(G)$.

[p. 169] (1) *We may assume that for each $e \in E(G)$, there is a leaf $t$ of $T$ with $E(\tau(t)) = \{e\}$ and $V(\tau(t))$ the set of ends of $e$, and hence that $E(\tau(t)) = \emptyset$ for each $t \in V(T)$ with valency $\geq 2$.*

For if for some $e$ there is no such $t$, we choose $t' \in V(T)$ with $e \in E(\tau(t'))$; we add a new vertex $t$ to $T$ adjacent only to $t'$; we remove $e$ from $\tau(t')$, and define $\tau(t)$ to be the hypergraph formed by $e$ and its ends. This provides a new tree-decomposition of $G$ of width $\omega(G)$. By continuing this process we may arrange that (1) holds.

(2) *We may assume that $|E(\tau(t))| = 1$ for each leaf $t$ of $T$.*

For by (1), $|E(\tau(t))| \leq 1$. If $E(\tau(t)) = \emptyset$ let $T'$ be obtained from $T$ by deleting $t$, and let $\tau'$ be the restriction of $\tau$ to $V(T')$; then since $G$ has no isolated vertices it follows that $(T', \tau')$ is a new tree-decomposition of $G$ of width $\omega(G)$ still satisfying (1). By continuing this process we may arrange that (2) holds.

(3) *We may assume that every vertex of $T$ has valency $\leq 3$.*

For if $t \in V(T)$ has valency $\geq 4$, we may choose a tree $T'$ and an edge $f$ of $T'$ such that $T$ is obtained from $T'$ by contracting $f$, and the two ends $t_1, t_2$ of $f$ both have valency less than the valency of $t$, and we define $\tau(t_1) = \tau(t_2) = \tau(t)$. The new tree-decomposition still has width $\omega(G)$ and still satisfies (1) and (2), and by repeating this process we may arrange that

(3) holds.

Now let $E(\tau(t)) = \{\sigma(t)\}$ for each leaf $t$ of $T$. Let $S$ be the tree obtained from $T$ by suppressing each vertex of valency 2. Then $(S, \sigma)$ is a branch-decomposition of $G$. For $f \in E(S)$, the order of $f$ in $(S, \sigma)$ is at most the number of vertices in $\tau(t)$, where $t$ is an end of $f$, and hence at most $\omega(G) + 1$. Thus $\beta(G) \leq \omega(G) + 1$, as required. □

Incidentally, both extremes of (5.1) can occur. For if $G = K_n$ (for some $n > 0$ divisible by 3) then $\omega(G) = \lfloor (3/2)\beta(G) \rfloor - 1$, by (4.4), since $\omega(G) = n-1$, while if $G$ is obtained from $K_{n,n}$ by deleting a perfect matching (for some $n \geq 4$) then it can be shown that $\omega(G) = n - 1$ and $\beta(G) = n$.

We deduce

(5.2) *For any hypergraph $G$, $\theta(G) \leq \omega(G) + 1 \leq (3/2)\theta(G)$.*

*Proof.* For from (5.1),

$$\max(\beta(G), \gamma(G)) \leq \omega(G) + 1 \leq \max(\frac{3}{2}\beta(G), \gamma(G), 1)$$

and from (4.3), $\max(\beta(G), \gamma(G)) = \theta(G)$ unless $\gamma(G) = 0$ and $V(G) \neq \emptyset$.

[p. 170] Moreover the proof of (2.5) shows that $\theta(G) \geq 1$ unless $V(G) = \emptyset$. Thus if $\gamma(G) \neq 0$ and hence $V(G) \neq \emptyset$, then

$$\begin{aligned}\theta(G) = \max(\beta(G), \gamma(G)) &\leq \omega(G) + 1 \leq \max(\frac{3}{2}\beta(G), \gamma(G), 1) \\ &= \frac{3}{2}\max(\beta(G), \gamma(G)) = \frac{3}{2}\theta(G),\end{aligned}$$

as required. If $\gamma(G) = 0$ and $V(G) \neq \emptyset$, then $\omega(G) = 0$ and $\theta(G) = 1$, and the result holds. Finally, if $V(G) = \emptyset$, then $\theta(G) = 0$ and $\omega(G) = -1$, and again the result holds. □

## 6. New Tangles from Old

The object of this section is to provide some operations on tangles. The simplest is the following. Let $\mathscr{T}$ be a tangle of order $\theta$ in a hypergraph $G$, let $1 \leq \theta' \leq \theta$, and let $\mathscr{T}'$ be the set of all members of $\mathscr{T}$ with order $< \theta'$. Then it is easy to see that $\mathscr{T}'$ is a tangle in $G$ of order $\theta'$; we call $\mathscr{T}'$ the *truncation* of $\mathscr{T}$ to order $\theta'$. We observe also that if $\mathscr{T}, \mathscr{T}'$ are tangles in $G$ then $\mathscr{T}' \subseteq \mathscr{T}$ if and only if $\mathscr{T}'$ is a truncation of $\mathscr{T}$.

For graphs $G$, a second construction extends a tangle in a minor of $G$ to a tangle in $G$, as follows.

(6.1) *Let $H$ be a minor of a graph $G$, and let $\mathscr{T}'$ be a tangle in $H$ of order $\theta \geq 2$. Let $\mathscr{T}$ be the set of all separations $(A, B)$ of $G$ of order $< \theta$*

*such that there exists* $(A', B') \in \mathscr{T}'$ *with* $E(A') = E(A) \cap E(H)$. *Then* $\mathscr{T}$ *is a tangle in* $G$ *of order* $\theta$.

*Proof.* We must verify the three axioms. First, let $(A, B)$ be a separation of $G$ of order $< \theta$. Then we may choose a separation $(A', B')$ of $H$ such that $E(A') = E(A) \cap E(H)$, and every vertex of $V(A' \cap B')$ is incident with an edge of $E(A')$ and with an edge of $E(B')$. Then $(A', B')$ has order at most the order of $(A, B)$ and so $< \theta$; thus, $\mathscr{T}'$ contains one of $(A', B')$, $(B', A')$, and so $\mathscr{T}$ contains one of $(A, B)$, $(B, A)$.

For the second axiom, suppose that $(A_i, B_i) \in \mathscr{T}$ $(1 \le i \le 3)$ with $A_1 \cup A_2 \cup A_3 = G$, and let $(A'_i, B'_i) \in \mathscr{T}'$ $(1 \le i \le 3)$ be the corresponding separations of $H$. Then $E(A'_1 \cup A'_2 \cup A'_3) = E(H)$, contrary to (2.3). Finally, it is clear from (2.7) that the third axiom holds. □

We call $\mathscr{T}$ in (6.1) the tangle in $G$ *induced by* $\mathscr{T}'$.

A third construction reverses this process. Let $G$ be a hypergraph and $W$ a set. We denote by $G/W$ the hypergraph $G'$ with vertex set $V(G) - W$ and edge set $E(G)$, in which $v \in V(G')$ and $e \in E(G')$ are incident if and only if they are incident in $G$. (This may produce edges with no ends.)

[p. 171] (6.2) *Let* $\mathscr{T}$ *be a tangle of order* $\theta$ *in a hypergraph* $G$, *and let* $W \subseteq V(G)$ *with* $|W| < \theta$. *Let* $\mathscr{T}'$ *be the set of all separations* $(A', B')$ *of* $G/W$ *such that there exists* $(A, B) \in \mathscr{T}$ *with* $W \subseteq V(A \cap B)$, $A/W = A'$, *and* $B/W = B'$. *Then* $\mathscr{T}'$ *is a tangle in* $G/W$ *of order* $\theta - |W|$.

*Proof.* Certainly every member of $\mathscr{T}'$ has order $< \theta - |W|$. For any separation $(A', B')$ of $G/W$ of order $< \theta - |W|$, there is a separation $(A, B)$ of $G$ of order $< \theta$ with $W \subseteq V(A \cap B)$, $A/W = A'$, and $B/W = B'$, and since $\mathscr{T}$ contains one of $(A, B)$, $(B, A)$, it follows that $\mathscr{T}'$ contains one of $(A', B')$, $(B', A')$. Thus the first axiom is satisfied.

For the second, suppose that $(A'_i, B'_i) \in \mathscr{T}'$ $(1 \le i \le 3)$. Choose $(A_i, B_i) \in \mathscr{T}$ with $W \subseteq V(A_i \cap B_i)$, $A_i/W = A'_i$, and $B_i/W = B'_i$ $(1 \le i \le 3)$. Since $A_1 \cup A_2 \cup A_3 \ne G$, it follows that $A'_1 \cup A'_2 \cup A'_3 \ne G/W$, and hence the second axiom holds.

For the third, let $(A', B') \in \mathscr{T}'$. Choose $(A, B) \in \mathscr{T}$ with $W \subseteq V(A \cap B)$, $A/W = A'$, and $B/W = B'$. Then $V(A) \ne V(G)$, and so $V(A') \ne V(G/W)$, as required. □

We denote the tangle $\mathscr{T}'$ of (6.2) by $\mathscr{T}/W$. We observe

(6.3) *Let* $\mathscr{T}, \theta, G, W$ *be as in (6.2), and let* $(A, B)$ *be a separation of* $G$. *Then* $(A/W, B/W) \in \mathscr{T}/W$ *if and only if* $(A, B) \in \mathscr{T}$ *and* $|V(A \cap B) - W| < \theta - |W|$.

*Proof.* Let $A^+$ be a subhypergraph of $G$ with $V(A^+) = V(A) \cup W$ and $E(A^+) = E(A)$, and define $B^+$ similarly. Then $(A^+, B^+)$ is a separation of

$G$, $W \subseteq V(A^+ \cap B^+)$, $A^+/W = A/W$, and $B^+/W = B/W$. By definition of $\mathscr{T}/W$, $(A^+, B^+) \in \mathscr{T}$ if and only if $(A/W, B/W) \in \mathscr{T}/W$. But by (2.9), $(A^+, B^+) \in \mathscr{T}$ if and only if $|V(A^+ \cap B^+)| < \theta$ and $(A, B) \in \mathscr{T}$. Since $|V(A^+ \cap B^+)| = |W| + |V(A \cap B) - W|$, the result follows. □

## 7. A Tangle in a Grid

Let $\theta \geq 2$ be an integer. Let $G$ be a simple graph with $V(G) = \{(i, j) : 1 \leq i, j \leq \theta\}$, where $(i, j)$ and $(i', j')$ are adjacent if $|i' - i| + |j' - j| = 1$. We call $G$ a *$\theta$-grid*. The object of this section is to prove the existence of a natural tangle of order $\theta$ in a $\theta$-grid.

Let $G$ be the $\theta$-grid defined above. For $1 \leq i \leq \theta$, let $P_i$ be the path of $G$ with vertex set $\{(i, j) : 1 \leq j \leq \theta\}$, and for $1 \leq j \leq \theta$, define $Q_j$ similarly. When $X \subseteq E(G)$, we define $\partial(X)$ to be the set of vertices $v \in XV(G)$[F3] such that $v$ is incident with an edge in $X$ and with an edge in $E(G) - X$.

(7.1) *If $X \subseteq E(G)$ and $|\partial(X)| < \theta$ then $X$ includes $E(P_i)$ for some $i$ $(1 \leq i \leq \theta)$ if and only if $X$ includes $E(Q_j)$ for some $j$ $(1 \leq j \leq \theta)$.*

[p. 172] *Proof.* Suppose that $E(P_i) \subseteq X$ for some $i$ $(1 \leq i \leq \theta)$. Then $V(Q_j)$ contains an end of an edge in $X$ for $1 \leq j \leq \theta$, since each $Q_j$ meets $P_i$. But not every $Q_j$ meets $\partial(X)$, since $|\partial(X)| < \theta$, and so for some $j$ $(1 \leq j \leq \theta)$, $E(Q_j) \subseteq X$, as required. □

If $X \subseteq E(G)$, we say that $X$ is *small* (in $G$) if $|\partial(X)| < \theta$ and $X$ includes $E(P_i)$ for no $i$ $(1 \leq i \leq \theta)$. The following is the main lemma used to obtain the required tangle, and we are grateful to D. Kleitman and M. Saks for finding the proof.

(7.2) *If $G$ is a $\theta$-grid and $X_1, X_2, X_3 \subseteq E(G)$ with $X_1 \cup X_2 \cup X_3 = E(G)$, then not all of $X_1, X_2, X_3$ are small in $G$.*

*Proof.* We proceed by induction on $\theta$. If $\theta = 2$ the result is trivial, and so we assume that $\theta \geq 3$ and that the result is true for $\theta - 1$. Let $P_1, \ldots, P_\theta, Q_1, \ldots, Q_\theta$ be as before.

If $E(Q_j) \subseteq X_1, X_2$, or $X_3$ for some $j$, the result is true by (7.1). Thus we may assume that each $E(Q_j)$ meets at least two of $X_1, X_2, X_3$, and in particular, without loss of generality, that

$$E(Q_\theta) \cap X_1 \neq \emptyset \neq E(Q_\theta) \cap X_2.$$

We suppose that all of $X_1, X_2, X_3$ are small. Thus, for $1 \leq j \leq \theta$ and $1 \leq k \leq 3$, if $E(Q_j)$ meets $X_k$, then $V(Q_j)$ meets $\partial(X_k)$. Moreover, if both

[F3] **Misprint.** Here $X$ is a set of edges, so its members are not vertices. With $v \in V(G)$, (7.1)–(7.3) hold as printed.

ends of $Q_j$ are incident with edges in $X_k$, then $|V(Q_j) \cap \partial(X_k)| \geq 2$. Now suppose that neither $E(P_1)$ nor $E(P_\theta)$ meets $X_3$. Then for $1 \leq j \leq \theta$ both ends of $Q_j$ are incident with edges in $X_1 \cup X_2$. From the above remarks, we deduce that

$$|V(Q_j) \cap \partial(X_1)| + |V(Q_j) \cap \partial(X_2)| \geq 2.$$

By summing over $j$, we find that $|\partial(X_1)| + |\partial(X_2)| \geq 2\theta$, a contradiction. Thus one of $E(P_1), E(P_\theta)$, say $E(P_\theta)$, meets $X_3$. Hence $E(P_\theta \cup Q_\theta)$ meets each of $X_1, X_2, X_3$ and hence $V(P_\theta \cup Q_\theta)$ meets each of $\partial(X_1), \partial(X_2), \partial(X_3)$.

Put $G' = G \setminus V(P_\theta \cup Q_\theta)$. Then $G'$ is a $(\theta-1)$-grid. Put $X'_k = X_k \cap E(G')$ $(1 \leq k \leq 3)$. Then $X'_1 \cup X'_2 \cup X'_3 = E(G')$. Let $\partial'$ be the $\partial$ function in $G'$.

Now

$$\partial'(X'_k) \subset \partial(X_k) \qquad (1 \leq k \leq 3)$$

since $V(P_\theta \cup Q_\theta)$ meets $\partial(X_k)$, and so

$$|\partial'(X'_k)| \leq \theta - 2 \qquad (1 \leq k \leq 3).$$

[p. 173] By our inductive hypothesis, one of $X'_1$, $X'_2$, $X'_3$ is not small in $G'$. By (7.1), we may choose $i', j'$ with $1 \leq i', j' \leq \theta - 1$, and $1 \leq k \leq 3$ such that

$$E((P_{i'} \cup Q_{j'}) \cap G') \subseteq X'_k.$$

If $k = 1$ or 2, then every $V(Q_j)$ contains an end of an edge in $X_k$ $(1 \leq j \leq \theta)$; for if $j = \theta$, this was shown earlier, and if $j < \theta$, then $V(Q_j)$ meets $V(P_{i'})$. Hence each $V(Q_j)$ meets $\partial(X_k)$, and so $|\partial(X_k)| \geq \theta$, a contradiction. Similarly, if $k = 3$, then every $V(P_i)$ meets $\partial(X_k)$, and again we have a contradiction. This completes the proof. □

From (7.2) we may infer the existence of the desired tangle. Given a $\theta$-grid $G$ with $P_1, \ldots, P_\theta, Q_1, \ldots, Q_\theta$ as before, let $\mathscr{T}$ be the set of all separations $(A, B)$ of $G$ of order $< \theta$ such that $E(A)$ is small.

(7.3) *$\mathscr{T}$ is a tangle in $G$ of order $\theta$.*

*Proof.* Let $(A, B)$ be a separation of $G$ of order $< \theta$. Suppose that neither $E(A)$ nor $E(B)$ is small. Choose $h, i$ with $1 \leq h, i \leq \theta$ such that $E(P_h) \subseteq E(A)$ and $E(P_i) \subseteq E(B)$. Thus $V(P_h) \subseteq V(A)$ and $V(P_i) \subseteq V(B)$. For $1 \leq j \leq \theta$, $\emptyset \neq V(Q_j \cap P_h) \subseteq V(Q_j \cap A)$, and similarly $V(Q_j \cap B) \neq \emptyset$, and so $V(Q_j \cap A \cap B) \neq \emptyset$ since $(A, B)$ is a separation. But then $|V(A \cap B)| \geq \theta$, a contradiction. Thus one of $E(A), E(B)$ is small, and so $\mathscr{T}$ satisfies the first axiom. That $\mathscr{T}$ is a tangle then follows from (7.2). □

The following was shown in [3].

(7.4) *For every $\theta \geq 2$ there exists $\phi \geq 0$ such that every graph with tree-width $\geq \phi$ has a $\theta$-grid minor.*

Since any graph with a $\theta$-grid minor has tree-width $\geq \theta$, one can say, roughly, that a graph has large tree-width if and only if it has a large grid minor. But (5.2) tells us that a graph has large tree-width if and only if it has a tangle of large order. One might therefore hope for a direct connection between tangles and grid minors, not via tree-width. The connection in one direction is easy, as follows. Let $H$ be a minor of $G$, isomorphic to a $\theta$-grid. Then the tangle in $H$ described in (7.3) induces a tangle $\mathscr{T}$ in $G$ of order $\theta$, by (6.1). A kind of converse is provided by the following strengthening of (7.4), proved in [7].

(7.5) *For every $\theta \geq 2$ there exists $\phi \geq 0$ such that for every graph $G$ and every tangle $\mathscr{T}$ in $G$ of order $\geq \phi$, the truncation of $\mathscr{T}$ to order $\theta$ is the tangle induced by some $\theta$-grid minor of $G$.*

## [p. 174] 8. Robust and Titanic Separations

The object of this section is to prove a technical lemma for use in a later paper. A separation $(A, B)$ of $G$ is *robust* if for every separation $(C, D)$ of $A$, one of the separations $(C, B \cup D)$, $(D, B \cup C)$ has order at least that of $(A, B)$. (Incidentally, Noga Alon (unpublished) has shown that deciding if a separation is robust is NP-complete.) We need the following lemma.

(8.1) *Let $(A, B)$ be a robust separation of $G$, and let $(C, D)$ be a separation of $G$. Then one of $(A \cup C, B \cap D)$, $(A \cup D, B \cap C)$ has order at most that of $(C, D)$.*

*Proof.* Now $(A \cap C, A \cap D)$ is a separation of $A$. Since $(A, B)$ is robust, we may assume (exchanging $C, D$ if necessary) that

$$|V((A \cap C) \cap (B \cup D))| = |V((A \cap C) \cap (B \cup (A \cap D)))| \geq |V(A \cap B)|.$$

But

$$\begin{aligned} &|V(A \cap B)| + |V(C \cap D)| \\ &\qquad = |V((A \cap C) \cap (B \cup D))| + |V((A \cup C) \cap (B \cap D))|, \end{aligned}$$

and the result follows. □

A separation $(A, B)$ of $G$ is *titanic* if for every triple $(X, Y, Z)$ of subhypergraphs of $A$ such that $A = X \cup Y \cup Z$ and $E(X)$, $E(Y)$, $E(Z)$ are mutually disjoint, we have either

$$|V((X \cup Y) \cap Z)| \geq |V((X \cup Y) \cap B)|$$

or

$$|V((Y \cup Z) \cap X)| \geq |V((Y \cup Z) \cap B)|$$

or

$$|V((Z \cup X) \cap Y)| \geq |V((Z \cup X) \cap B)|.$$

(8.2) *Every titanic separation is robust.*

*Proof.* Let $(A, B)$ be a titanic separation, and let $(C, D)$ be a separation of $A$. Put $X = C$, $Y = D$, and let $Z$ be the hypergraph with $V(Z) = E(Z) = \emptyset$. Since $(A, B)$ is titanic, we deduce that either $0 \geq |V(A \cap B)|$ or $|V(C \cap D)| \geq |V(B \cap D)|$ or $|V(C \cap D)| \geq |V(B \cap C)|$. If $V(A \cap B) = \emptyset$ then $(A, B)$ is robust. Thus, by symmetry, we may assume that $|V(B \cap C)| \leq |V(C \cap D)|$. But

$$|V(A \cap B)| = |V(B \cap C)| + |V(B \cap D) - V(C)|$$

[p. 175] and

$$|V((B \cup C) \cap D)| = |V(C \cap D)| + |V(B \cap D) - V(C)|,$$

and so $|V(A \cap B)| \leq |V((B \cup C) \cap D)|$, as required. □

The main result of this section is another way to construct new tangles from old, the following.

(8.3) *Let $(C, D)$ be a separation of a hypergraph $G$, and let $(C', D)$ be a titanic separation of a hypergraph $G'$, with $V(C \cap D) = V(C' \cap D)$. Let $\mathscr{T}$ be a tangle in $G$ of order $\theta \geq 2$ with $(C, D) \in \mathscr{T}$. Let $\mathscr{T}'$ be the set of all separations $(A', B')$ of $G'$ of order $< \theta$ such that there exists $(A, B) \in \mathscr{T}$ with $E(A \cap D) = E(A' \cap D)$. Then $\mathscr{T}'$ is a tangle in $G'$ of order $\theta$.*

*Proof.* We verify the hypotheses of (4.5). For the first axiom, let $(A', B')$ be a separation of $G'$ of order $< \theta$. Since $(C', D)$ is robust by (8.2), we may assume by (8.1) (exchanging $A', B'$ if necessary) that $(A' \cap D, B' \cup C')$ has order at most that of $(A', B')$. Now $(A' \cap D, (B' \cap D) \cup C)$ is a separation of $G$ with the same order as $(A' \cap D, B' \cup C')$, since $B' \cup C' = (B' \cap D) \cup C'$ and

$$(A' \cap D) \cap C = A' \cap (D \cap C) = A' \cap (D \cap C') = (A' \cap D) \cap C'.$$

Hence $(A' \cap D, (B' \cap D) \cup C)$ has order $< \theta$ and so $\mathscr{T}$ contains one of $(A' \cap D, (B' \cap D) \cup C)$, $((B' \cap D) \cup C, A' \cap D)$. If the first, then $(A', B') \in \mathscr{T}'$, while if the second, then since $E(((B' \cap D) \cup C) \cap D) = E(B' \cap D)$, it follows that $\mathscr{T}'$ contains $(B', A')$. This verifies that $\mathscr{T}'$ satisfies the first axiom.

For (4.5) (i), suppose that $(A'_1, B'_1), (A'_2, B'_2) \in \mathscr{T}'$. Choose $(A_i, B_i) \in \mathscr{T}$ with $E(A_i \cap D) = E(A'_i \cap D)$ $(i = 1, 2)$. Since $(C, D) \in \mathscr{T}$, $E(C \cup A_1 \cup A_2) \neq E(G)$ by (2.3), and so $E(D) \not\subseteq E(A_1 \cup A_2)$. Hence $E(D) \not\subseteq E(A'_1 \cup A'_2)$, and so $A'_1 \cup A'_2 \neq G'$, and $B'_1 \not\subseteq A'_2$, as required.

For (4.5) (ii), suppose that $A'_1$, $A'_2$, $A'_3$ are mutually edge-disjoint subhypergraphs of $G'$ with union $G'$, and $(A'_i, B'_i) \in \mathscr{T}'$ for $i = 1, 2, 3$, where $B'_1 = A'_2 \cup A'_3$, $B'_2 = A'_3 \cup A'_1$, $B'_3 = A'_1 \cup A'_2$. Choose $(A_i, B_i) \in \mathscr{T}$ with $E(A_i \cap D) = E(A'_i \cap D)$ $(1 \le i \le 3)$. Let $F_i = A'_i \cap C'$ $(1 \le i \le 3)$. Then $F_1 \cup F_2 \cup F_3 = C'$, and since $(C', D)$ is titanic we may renumber so that

$$|V((F_2 \cup F_3) \cap F_1)| \ge |V((F_2 \cup F_3) \cap D)|;$$

that is,

$$|V(B'_1 \cap C' \cap A'_1)| \ge |V(B'_1 \cap C' \cap D)|.$$

[p. 176] Now $V(A'_1 \cup C') = V(C') \cup (V(A'_1) - V(C'))$, and so

$$\begin{aligned} &|V((A'_1 \cup C') \cap (B'_1 \cap D))| \\ &\qquad = |V(B'_1 \cap C' \cap D)| + |(V(A'_1) - V(C')) \cap V(B'_1 \cap D)|. \end{aligned}$$

Moreover, since $V(A'_1 \cap B'_1) - V(C') = (V(A'_1) - V(C')) \cap V(B'_1 \cap D)$, it follows that

$$|V(A'_1 \cap B'_1)| = |V(B'_1 \cap C' \cap A'_1)| + |(V(A'_1) - V(C')) \cap V(B'_1 \cap D)|.$$

We deduce that $(A'_1 \cup C', B'_1 \cap D)$ has order at most that of $(A'_1, B'_1)$ and hence $< \theta$. It follows that $((A'_1 \cap D) \cup C, B'_1 \cap D)$ is a separation of $G$ of order $< \theta$, and so $\mathscr{T}$ contains one of $(B'_1 \cap D, (A'_1 \cap D) \cup C)$, $((A'_1 \cap D) \cup C, B'_1 \cap D)$. The first is impossible by (2.3), since $(C, D), (A_1, B_1) \in \mathscr{T}$ and

$$E((B'_1 \cap D) \cup C \cup A_1) = E(G).$$

The second is impossible by (2.3), since $(A_2, B_2), (A_3, B_3) \in \mathscr{T}$ and

$$E((A'_1 \cap D) \cup C \cup A_2 \cup A_3) = E(G).$$

This contradiction completes the verification of (4.5) (ii). Thus, from (4.5), we deduce that $\mathscr{T}'$ satisfies the second axiom.

To verify the third axiom, we verify the hypothesis of (2.7). Let $e \in E(G')$ with size $< \theta$, and let $K_e$ be as in (2.7). If $e \in E(D)$, then since $(K_e, G \setminus e) \in \mathscr{T}$ by (2.7) applied to $G$, $\mathscr{T}$, it follows from the definition of $\mathscr{T}'$ that $(K_e, G' \setminus e) \in \mathscr{T}'$. If $e \in E(C')$, then since $(C, D) \in \mathscr{T}$ and $E(C \cap D) = E(K_e \cap D)$, it again follows that $(K_e, G' \setminus e) \in \mathscr{T}'$ from the definition of $\mathscr{T}'$. Thus, from (2.7), we deduce that $\mathscr{T}'$ satisfies the third axiom, as required. □

As an application, we observe

(8.4) *Let $\mathscr{T}$ be a tangle of order $\theta \geq 2$ in a hypergraph $G$, and let $e \in E(G)$ with at most one end. Let $\mathscr{T}'$ be the set of all separations $(A', B')$ of $G \setminus e$ of order $< \theta$ such that there exists $(A, B) \in \mathscr{T}$ with $E(A \cap (G \setminus e)) = E(A')$. Then $\mathscr{T}'$ is a tangle in $G \setminus e$ of order $\theta$.*

*Proof.* Let $C$ be the subhypergraph of $G$ formed by $e$ and its ends and let $C' = C \setminus e$ and $D = G \setminus e$. Then $(C, D) \in \mathscr{T}$ and $(C', D)$ is titanic, as is easily seen, and the result follows from (8.3). □

Thus, if we delete all edges of $G$ with $\leq 1$ end, we do not change its tangle number. (This holds even for tangle number $\leq 1$, as is easily seen.) (8.4) has the following consequence.

[p. 177] (8.5) *Let $\mathscr{T}$ be a tangle in a graph $G$ of order $\theta \geq 1$. Let $W \subseteq V(G)$ with $|W| < \theta$. Let $\mathscr{T}'$ be the set of all separations $(A', B')$ of $G \setminus W$ of order $< \theta - |W|$ such that there exists $(A, B) \in \mathscr{T}$ with $W \subseteq V(A \cap B)$ and $A \setminus W = A'$, $B \setminus W = B'$. Then $\mathscr{T}'$ is a tangle in $G \setminus W$ of order $\theta - |W|$.*

*Proof.* Since $|W| < \theta$, the result is obvious when $\theta = 1$, and so we may assume that $\theta \geq 2$. Now $G \setminus W$ is obtained from $G/W$ by deleting edges with at most one end, and $\mathscr{T}'$ is obtained from $\mathscr{T}/W$ by repeating the operation of (8.4). The result follows. □

## 9. Laminar Separations

We have seen in (5.2) that the tangles of large order are obstructions to the existence of tree-decompositions of small width. Our next result is a counterpart of this, that there is a tree-decomposition into pieces which correspond to the tangles.

Let $(A_1, B_1)$, $(A_2, B_2)$ be separations of a hypergraph $G$. We say these separations *cross* unless either $A_1 \subseteq A_2$ and $B_2 \subseteq B_1$, or $A_1 \subseteq B_2$ and $A_2 \subseteq B_1$, or $B_1 \subseteq A_2$ and $B_2 \subseteq A_1$, or $B_1 \subseteq B_2$ and $A_2 \subseteq A_1$. A set of separations is *laminar* if no two of its members cross.

Let $(T, \tau)$ be a tree-decomposition of a hypergraph $G$. For each $e \in E(T)$, let $T_1$, $T_2$ be the components of $T \setminus e$ and let

$$G_i^e = \bigcup(\tau(t) : t \in V(T_i)) \qquad (i = 1, 2).$$

Then $(G_1^e, G_2^e)$ is a separation of $G$, and we call $(G_1^e, G_2^e)$ and $(G_2^e, G_1^e)$ the separations *made by* $e$ (under the given tree-decomposition).

(9.1) *If $(T, \tau)$ is a tree-decomposition of $G$, then the set of all separations of $G$ made by edges of $T$ is laminar. Conversely, if $\{(A_i, B_i) : 1 \leq i \leq k\}$ is a laminar set of separations of $G$, there is a tree-decomposition $(T, \tau)$ of $G$ such that*

(i) *for* $1 \le i \le k$, $(A_i, B_i)$ *is made by a unique edge of* $T$

(ii) *for each edge* $e$ *of* $T$*, at least one of the two separations made by* $e$ *equals* $(A_i, B_i)$ *for some* $i$ $(1 \le i \le k)$*.*

The proof is easy ~~and is left to the reader~~, and we give it now.

For an edge $e$ of $T$ with ends $u, v$, let $G^e_u$ be the union of the subhypergraphs $\tau(t)$ over the vertices $t$ of the component of $T \setminus e$ that contains $u$. Thus $(G^e_u, G^e_v)$ and $(G^e_v, G^e_u)$ are the separations made by $e$. (They are separations, since $G^e_u \cup G^e_v = G$ by (i) of the definition of tree-decomposition, and no edge of $G$ is in both, by (ii).) If we replace one of two separations by its reverse, the four conditions in the definition of crossing are permuted. Hence two separations cross if and only if they cross after one or both of them are reversed.

For the first statement, let $e_1, e_2 \in E(T)$. If $e_1 = e_2$, the separations made by $e_1$ are $(A, B)$ and $(B, A)$ for some $A, B$, and two separations each equal to $(A, B)$ or $(B, A)$ do not cross: if they are equal the first condition holds, and otherwise the second holds. So we may assume that $e_1 \ne e_2$. Let $e_i$ have ends $u_i, v_i$ $(i = 1, 2)$. Since $e_2$ is an edge of $T \setminus e_1$, its ends are in the same component of $T \setminus e_1$, and by exchanging the names $u_1, v_1$ we may assume that this is the component that contains $v_1$. Similarly, by exchanging the names $u_2, v_2$, which keeps this property, we may assume that $u_1$ and $v_1$ are in the component of $T \setminus e_2$ that contains $u_2$. The component of $T \setminus e_1$ that contains $u_1$ does not contain $e_2$, since both ends of $e_2$ are in the other component, and so it is a connected subgraph of $T \setminus e_2$. It contains $u_1$, and so it is contained in the component of $T \setminus e_2$ that contains $u_2$. Hence $G^{e_1}_{u_1} \subseteq G^{e_2}_{u_2}$. Similarly the component of $T \setminus e_2$ that contains $v_2$ does not contain $e_1$, since both ends of $e_1$ are in the component that contains $u_2$. So it is a connected subgraph of $T \setminus e_1$, and it contains $v_2$, which is in the component of $T \setminus e_1$ that contains $v_1$. Hence $G^{e_2}_{v_2} \subseteq G^{e_1}_{v_1}$. So $(G^{e_1}_{u_1}, G^{e_1}_{v_1})$ and $(G^{e_2}_{u_2}, G^{e_2}_{v_2})$ satisfy the first condition and do not cross. The other separations made by $e_1$ and $e_2$ are their reverses, and so no two of the separations made by edges of $T$ cross.

For the converse, let $\mathscr{S}$ be the set of the separations $(A_i, B_i)$ and $(B_i, A_i)$ $(1 \le i \le k)$. It is laminar, by the remark on reversal. We prove by induction on $|\mathscr{S}|$ that if $\mathscr{S}$ is a laminar set of separations of $G$ that contains the reverse of each of its members, then there is a tree-decomposition $(T, \tau)$ of $G$ such that the separations made by edges of $T$ are exactly the members of $\mathscr{S}$, and each member of $\mathscr{S}$ is made by a unique edge of $T$. This proves (i) and (ii), since the two separations made by an edge of $T$ are the reverses of each other. If $\mathscr{S} = \emptyset$, let $T$ have one vertex $t$ and put $\tau(t) = G$. So we may

assume that $\mathscr{S} \neq \emptyset$. For a separation $(A, B)$ of $G$ let

$$s(A, B) = |V(A)| + |E(A)| + |V(G) - V(B)| + |E(G) - E(B)|,$$

and choose $(A_0, B_0) \in \mathscr{S}$ with $s(A_0, B_0)$ minimum. Let $\mathscr{S}'$ be obtained from $\mathscr{S}$ by removing $(A_0, B_0)$ and $(B_0, A_0)$. Then $\mathscr{S}'$ is laminar and contains the reverse of each of its members, and so from our inductive hypothesis there is a tree-decomposition $(T', \tau')$ of $G$ for $\mathscr{S}'$ as above. For an edge $e$ of $T'$ with ends $u, v$ we define $G'^e_u$ for $(T', \tau')$ as we defined $G^e_u$.

(1) *For each $(C, D) \in \mathscr{S}'$, either $A_0 \subseteq C$ and $D \subseteq B_0$, or $A_0 \subseteq D$ and $C \subseteq B_0$.*

For $(A_0, B_0)$ and $(C, D)$ do not cross, and so one of the four conditions holds. The first two are the two outcomes of (1). Suppose that $B_0 \subseteq C$ and $D \subseteq A_0$. Then $(D, C) \in \mathscr{S}$, and $s(D, C) \leq s(A_0, B_0)$ with equality only if $D = A_0$ and $C = B_0$, since $D \subseteq A_0$ and $B_0 \subseteq C$. By the choice of $(A_0, B_0)$ equality holds, and so $(C, D) = (B_0, A_0) \notin \mathscr{S}'$, a contradiction. Similarly, if $B_0 \subseteq D$ and $C \subseteq A_0$, then $(C, D) = (A_0, B_0)$, again a contradiction. This proves (1).

Call an ordered pair $(u, w)$ of adjacent vertices of $T'$ *good* if $A_0 \subseteq G'^e_u$ and $G'^e_w \subseteq B_0$, where $e$ is the edge with ends $u, w$. Since $(G'^e_u, G'^e_w) \in \mathscr{S}'$, (1) implies that for every edge $e$ of $T'$ with ends $u, w$, at least one of $(u, w)$, $(w, u)$ is good.

(2) *There is a vertex $u_0$ of $T'$ such that $(u_0, w)$ is good for every neighbour $w$ of $u_0$.*

For suppose not. Then each vertex $v$ of $T'$ has a neighbour $u$ such that $(v, u)$ is not good, and then $(u, v)$ is good. Starting from any vertex and repeating, we obtain an infinite walk $v_0 v_1 v_2 \ldots$ of $T'$ such that for each $i$, $(v_i, v_{i+1})$ is not good and $(v_{i+1}, v_i)$ is good. Then $v_{i+2} \neq v_i$, since $(v_{i+1}, v_{i+2})$ is not good and $(v_{i+1}, v_i)$ is good. A walk in a tree that never goes back along the edge it has just used is a path, and so $T'$ has arbitrarily long paths, which is impossible. This proves (2).

(3) *$\tau'(t) \subseteq B_0$ for every vertex $t \neq u_0$ of $T'$, and $A_0 \subseteq \tau'(u_0)$.*

Let $t \neq u_0$, and let $w$ be the neighbour of $u_0$ on the path of $T'$ from $u_0$ to $t$, and $e$ the edge with ends $u_0, w$. Then $t$ is in the component of $T' \setminus e$ that contains $w$, and so $\tau'(t) \subseteq G'^e_w \subseteq B_0$ by (2). For the second statement, let $x \in V(A_0)$. Then $x \in V(\tau'(t))$ for some $t$. If $t \neq u_0$, define $w$ and $e$ as before. Since $A_0 \subseteq G'^e_{u_0}$ by (2), $x \in V(\tau'(t_1))$ for some vertex $t_1$ of the component of $T' \setminus e$ that contains $u_0$. Then $u_0$ is on the path of $T'$ between $t_1$ and $t$, and so $x \in V(\tau'(u_0))$. Now let $f \in E(A_0)$. Then $f \in E(\tau'(t))$ for some $t$, and $t = u_0$, since otherwise $f \in E(B_0)$ by the first statement, which is impossible since $E(A_0 \cap B_0) = \emptyset$. This proves (3).

Now let $T$ be obtained from $T'$ by adding a new vertex $t_0$ and an edge $e_0$ joining $t_0$ and $u_0$. Let $\tau(t_0) = A_0$, $\tau(u_0) = \tau'(u_0) \cap B_0$, and $\tau(t) = \tau'(t)$ for every other vertex $t$. Since $\tau'(u_0) \subseteq A_0 \cup B_0$ and $A_0 \subseteq \tau'(u_0)$ by (3),

$$\tau(t_0) \cup \tau(u_0) = A_0 \cup (\tau'(u_0) \cap B_0) = \tau'(u_0).$$

We check that $(T, \tau)$ is a tree-decomposition of $G$. First, $T$ is a tree, and (i) holds by (i) for $(T', \tau')$ and the equation above. For (ii), $\tau(t_0) = A_0$ has no edge in common with $\tau(u_0)$ or with any $\tau(t) = \tau'(t)$ $(t \neq t_0, u_0)$, since these are subhypergraphs of $B_0$ by (3) and $E(A_0 \cap B_0) = \emptyset$. The other pairs satisfy (ii) since $\tau(t) \subseteq \tau'(t)$ for every vertex $t$ of $T'$. For (iii), let $t'$ be on the path of $T$ between $t$ and $t''$, where $t' \neq t, t''$ (otherwise (iii) is trivial). Then $t' \neq t_0$, since $t_0$ has valency 1. If $t_0 \notin \{t, t''\}$, then $\tau(t) \cap \tau(t'') \subseteq \tau'(t')$ by (iii) for $(T', \tau')$. If $t_0 \in \{t, t''\}$, say $t = t_0$, then $t'$ is on the path of $T'$ between $u_0$ and $t''$. Since $A_0 \subseteq \tau'(u_0)$, it follows that $\tau(t_0) \cap \tau(t'') \subseteq \tau'(u_0) \cap \tau'(t'') \subseteq \tau'(t')$. So in each case $\tau(t) \cap \tau(t'') \subseteq \tau'(t')$. This proves (iii) when $t' \neq u_0$. When $t' = u_0$, neither $t$ nor $t''$ is $u_0$, and one of them is not $t_0$. That one is a vertex $t_1 \neq u_0$ of $T'$, and $\tau(t_1) \subseteq B_0$ by (3). Hence $\tau(t) \cap \tau(t'') \subseteq \tau'(u_0) \cap B_0 = \tau(u_0)$. So $(T, \tau)$ is a tree-decomposition of $G$.

The separations made by $e_0$ are $(A_0, B_0)$ and $(B_0, A_0)$. For the component of $T \setminus e_0$ that contains $t_0$ has only the vertex $t_0$, and $\tau(t_0) = A_0$. The union of $\tau(t)$ over the other vertices is a subhypergraph of $B_0$ by (3). It also contains every vertex and edge of $B_0$, since each of them is in some $\tau'(t)$, and if $t = u_0$ it is in $\tau'(u_0) \cap B_0 = \tau(u_0)$. So this union is $B_0$. Now let $e$ be an edge of $T'$. Then $t_0$ and $u_0$ are in the same component of $T \setminus e$, and $\tau(t_0) \cup \tau(u_0) = \tau'(u_0)$. So $e$ makes the same separations under $(T, \tau)$ as under $(T', \tau')$. Hence the separations made by edges of $T$ are the members of $\mathscr{S}' \cup \{(A_0, B_0), (B_0, A_0)\} = \mathscr{S}$. Each member of $\mathscr{S}'$ is made by a unique edge of $T'$, and not by $e_0$. The separations $(A_0, B_0)$ and $(B_0, A_0)$ are made by $e_0$, and by no edge of $T'$, since they are not in $\mathscr{S}'$. This completes the induction, and so proves (9.1).[F4]

We wish to arrange a "tie-breaking" mechanism so that no two distinct separations are counted as having the same order (except for reversal). A *tie-breaker* $\lambda$ in a hypergraph $G$ is a function from the set of all separations of $G$ into some linearly ordered set $(\Lambda, <)$, satisfying certain axioms given below. For each separation $(A, B)$, $\lambda(A, B)$ is called the $\lambda$-*order* of $(A, B)$, [p. 178] and, if $(A, B)$, $(C, D)$ are separations, we say that $(A, B)$ has *smaller*

[F4] **Missing proof.** The paper leaves this proof to the reader. The argument above is the one used in the formalization.

*$\lambda$-order than* $(C, D)$ if $\lambda(A, B) < \lambda(C, D)$. The tie-breaker $\lambda$ must satisfy the following conditions:

(i) if $(A, B)$, $(C, D)$ are separations of $G$, they have the same $\lambda$-order if and only if $(A, B) = (C, D)$ or $(A, B) = (D, C)$

(ii) if $(A, B)$, $(C, D)$ are separations of $G$, then either $(A \cup C, B \cap D)$ has $\lambda$-order at most that of $(A, B)$ or $(A \cap C, B \cup D)$ has $\lambda$-order smaller than that of $(C, D)$

(iii) if $(A, B)$, $(C, D)$ are separations of $G$ and $(A, B)$ has smaller order than $(C, D)$, then $(A, B)$ has smaller $\lambda$-order than $(C, D)$.

We refer to these as the *first*, *second*, and *third tie-breaker axioms*.

(9.2) *In every hypergraph there is a tie-breaker.*

*Proof.* Let $(\Lambda, <)$ be the set of all triples of real numbers, ordered lexicographically; thus, $(a, b, c) < (a', b', c')$ if $a < a'$, or $a = a'$ and $b < b'$, or $a = a'$ and $b = b'$ and $c < c'$. For any hypergraph $G$, let $L(G) = V(G) \cup E(G)$. Let $G$ be a hypergraph. Choose a function $\mu$ from $L(G) \times L(G)$ into the set of positive real numbers such that

(i) $\mu(x, y) = \mu(y, x)$ for all $x, y \in L(G)$, and

(ii) for every choice of rationals $\alpha(x, y)$ $(x, y \in L(G))$ such that $\sum_{x,y} \alpha(x, y)\mu(x, y) = 0$, we have $\alpha(x, y) = -\alpha(y, x)$ for all $x, y \in L(G)$.

For each separation $(A, B)$ of $G$, define $\lambda(A, B) = (N_1, N_2, N_3)$, where

$$N_1 = |V(A \cap B)|$$
$$N_2 = \sum(\mu(x, x) : x \in V(A \cap B))$$
$$N_3 = \sum(\mu(x, y) : x \in L(A) - L(B), y \in L(B) - L(A)).$$

(1) *If $(A, B)$ and $(A', B')$ are separations of $G$ with the same $\lambda$-order then $(A', B') = (A, B)$ or $(B, A)$.*

For let $(A, B)$ have $\lambda$-order $(N_1, N_2, N_3)$, and let $(A', B')$ have $\lambda$-order $(N_1', N_2', N_3')$. Let $V(A \cap B) = Z$, $L(A) - L(B) = X$, $L(B) - L(A) = Y$, and define $Z'$, $X'$, $Y'$ similarly. Then $(X, Y, Z)$, $(X', Y', Z')$ are partitions of $L(G)$, and we must show that $Z' = Z$ and that $(X', Y') = (X, Y)$ or $(Y, X)$. Now since $N_2 = N_2'$,

$$\sum_{x \in Z} \mu(x, x) = \sum_{x \in Z'} \mu(x, x),$$

[p. 179] and so $Z = Z'$ from (ii) above. Moreover, since $N_3 = N_3'$,

$$\sum(\mu(x, y) : x \in X, y \in Y) = \sum(\mu(x, y) : x \in X', y \in Y').$$

Hence

$$\{\{x, y\} : x \in X, y \in Y\} = \{\{x, y\} : x \in X', y \in Y'\},$$

and the claim follows.

(2) *Let $(A,B)$, $(C,D)$ be separations of $G$. Then so are $(A\cup C, B\cap D)$, $(A\cap C, B\cup D)$, and the sum of their $\lambda$-orders is at most the sum of the $\lambda$-orders of $(A,B)$, $(C,D)$.*

This follows by comparing (for each $x,y\in L(G)$) the number of occurrences of $\mu(x,y)$ and $\mu(y,x)$ in the expressions for the $\lambda$-orders of $(A,B)$ and $(C,D)$ with the corresponding numbers for the other two separations.

From (1) and (2), it follows that the first and second tie-breaker axioms are satisfied, and clearly so is the third, as required. □

The following strengthening of the second axiom is sometimes useful.

(9.3) *Let $\lambda$ be a tie-breaker in a hypergraph $G$, and let $(A,B)$, $(C,D)$ be separations of $G$. Then either* (i) *$(A\cup C, B\cap D)$ has smaller $\lambda$-order than $(A,B)$, or* (ii) *$(A\cap C, B\cup D)$ has smaller $\lambda$-order than $(C,D)$, or* (iii) *$C\subseteq A$ and $B\subseteq D$, or* (iv) *$B=C=G$ and $A=D$ and $E(A)=\emptyset$.*

*Proof.* Since we may assume that (ii) is false, it follows from the second axiom that $(A\cup C, B\cap D)$ has $\lambda$-order at most that of $(A,B)$, and we may assume that equality holds, for otherwise (i) holds. Thus, by the first axiom, $(A\cup C, B\cap D) = (A,B)$ or $(B,A)$. If $(A\cup C, B\cap D)=(A,B)$ then $C\subseteq A$ and $B\subseteq D$ and (iii) holds, and so we may assume that $(A\cup C, B\cap D)=(B,A)$. Hence $A\cup C = B$ and $B\cap D = A$. In particular, $A\subseteq B$, and since $A\cup B = G$, it follows that $B=G$, and $A=D$ since $B\cap D = A$.

By the second axiom applied to $(D,C)$, $(B,A)$, we deduce that either $(B\cup D, A\cap C)$ has $\lambda$-order at most that of $(D,C)$ or $(B\cap D, A\cup C)$ has $\lambda$-order less than $(B,A)$. In the second case, (i) holds, and if strict inequality holds in the first case, then (ii) holds. Thus we may assume that $(B\cup D, A\cap C)$ has the same $\lambda$-order as $(D,C)$, and so $(B\cup D, A\cap C) = (D,C)$ or $(C,D)$, by the first axiom. In the first case, $B\subseteq D$ and $C\subseteq A$, and (iii) holds, and so we may assume that $(B\cup D, A\cap C)=(C,D)$; that is, $C=G$ and $A=D$. Since $B=G$, it follows that (iv) holds. □

[p. 180] Given a tie-breaker $\lambda$, a separation $(A,B)$ of $G$ is *$\lambda$-robust* if for every separation $(C,D)$ of $A$, one of $(C, B\cup D)$, $(D, B\cup C)$ has $\lambda$-order at least the $\lambda$-order of $(A,B)$. Clearly a $\lambda$-robust separation is robust. The separation $(A,B)$ is *doubly $\lambda$-robust* if both $(A,B)$ and $(B,A)$ are $\lambda$-robust.

(9.4) *Let $(A,B)$, $(C,D)$ be doubly $\lambda$-robust separations of $G$. Then $(A,B)$ and $(C,D)$ do not cross.*

*Proof.* By the symmetry, we may assume that of the four separations $(A\cap C, B\cup D)$, $(A\cap D, B\cup C)$, $(B\cap C, A\cup D)$, $(B\cap D, A\cup C)$, the first has smallest $\lambda$-order. Since $(C\cap A, D\cap A)$ is a separation of $A$ and $(A,B)$

is $\lambda$-robust, one of

$$(C \cap A, (D \cap A) \cup B), \qquad (D \cap A, (C \cap A) \cup B)$$

has $\lambda$-order at least that of $(A, B)$. These separations are $(A \cap C, B \cup D)$ and $(A \cap D, B \cup C)$, respectively, and so, in view of the assumption in the first sentence of this proof, $(A \cap D, B \cup C)$ has $\lambda$-order at least that of $(A, B)$. Similarly, $(B \cap C, A \cup D)$ has $\lambda$-order at least that of $(C, D)$. By (9.3) applied to $(B, A)$, $(C, D)$, we deduce that either $C \subseteq B$ and $A \subseteq D$, or $A = C = G$ and $B = D$, and in either case $(A, B)$, $(C, D)$ do not cross. □

## 10. Tangle Tree-Decompositions

Let $\mathscr{T}_1$, $\mathscr{T}_2$ be tangles in a graph $G$. They are *indistinguishable* if one is a truncation of the other, that is, either $\mathscr{T}_1 \subseteq \mathscr{T}_2$ or $\mathscr{T}_2 \subseteq \mathscr{T}_1$, and otherwise they are *distinguishable*. A separation $(A, B)$ of $G$ *distinguishes* $\mathscr{T}_1$ *from* $\mathscr{T}_2$ if $(A, B) \in \mathscr{T}_1$ and $(B, A) \in \mathscr{T}_2$.

(10.1) *Either there is a separation of $G$ which distinguishes $\mathscr{T}_1$ from $\mathscr{T}_2$ or $\mathscr{T}_1$, $\mathscr{T}_2$ are indistinguishable and not both.*

*Proof.* Since there is a separation distinguishing $\mathscr{T}_1$ from $\mathscr{T}_2$ if and only if there is one distinguishing $\mathscr{T}_2$ from $\mathscr{T}_1$, we may assume that $\mathscr{T}_2$ has order at least that of $\mathscr{T}_1$. Then

$$\begin{aligned}
&\mathscr{T}_1 \text{ and } \mathscr{T}_2 \text{ are distinguishable} \\
&\quad \Leftrightarrow \mathscr{T}_1 \not\subseteq \mathscr{T}_2 \\
&\quad \Leftrightarrow \text{there exists } (A, B) \in \mathscr{T}_1 \text{ with } (A, B) \notin \mathscr{T}_2 \\
&\quad \Leftrightarrow \text{there exists } (A, B) \in \mathscr{T}_1 \text{ with } (B, A) \in \mathscr{T}_2 \\
&\quad \Leftrightarrow \text{there is a separation distinguishing } \mathscr{T}_1 \text{ from } \mathscr{T}_2,
\end{aligned}$$

as required. □

[p. 181] Given a tie-breaker $\lambda$, a separation $(A, B)$ which distinguishes $\mathscr{T}_1$ from $\mathscr{T}_2$ is a $(\mathscr{T}_1, \mathscr{T}_2)$-*distinction* if it has minimum $\lambda$-order of all separations which distinguish $\mathscr{T}_1$ from $\mathscr{T}_2$. From the first tie-breaker axiom, $(A, B)$ is unique, and we may speak of *the* $(\mathscr{T}_1, \mathscr{T}_2)$-distinction. (Of course, different choices of the tie-breaker $\lambda$ result in different $(\mathscr{T}_1, \mathscr{T}_2)$-distinctions in general.) There is a $(\mathscr{T}_1, \mathscr{T}_2)$-distinction if and only if $\mathscr{T}_1, \mathscr{T}_2$ are distinguishable.

(10.2) *If $\mathscr{T}_1, \mathscr{T}_2$ are distinguishable tangles in $G$, the $(\mathscr{T}_1, \mathscr{T}_2)$-distinction is doubly $\lambda$-robust.*

*Proof.* Let $(A, B)$ be the $(\mathscr{T}_1, \mathscr{T}_2)$-distinction. Since $(B, A)$ is the $(\mathscr{T}_2, \mathscr{T}_1)$-distinction, it suffices to show that $(A, B)$ is $\lambda$-robust. Let $(C, D)$ be a separation of $A$, and suppose that both $(C, B \cup D)$ and $(D, B \cup C)$ have $\lambda$-order strictly smaller than that of $(A, B)$. Then $(C, B \cup D)$, $(D, B \cup C)$ have order at most that of $(A, B)$ and hence less than the orders of $\mathscr{T}_1$ and $\mathscr{T}_2$. Since $(A, B) \in \mathscr{T}_1$ it follows that $(C, B \cup D) \in \mathscr{T}_1$ and $(D, B \cup C) \in \mathscr{T}_1$. Since $(A, B)$ is the $(\mathscr{T}_1, \mathscr{T}_2)$-distinction it follows that $(B \cup D, C) \notin \mathscr{T}_2$ and $(B \cup C, D) \notin \mathscr{T}_2$, and hence $(C, B \cup D)$, $(D, B \cup C) \in \mathscr{T}_2$. But $(B, A) \in \mathscr{T}_2$, and $B \cup C \cup D = G$, contrary to the second tangle axiom. Thus one of $(C, B \cup D)$, $(D, B \cup C)$ has $\lambda$-order at least that of $(A, B)$, and hence $(A, B)$ is $\lambda$-robust, as required. □

(10.3) *Let $\mathscr{T}_1, \dots, \mathscr{T}_n$ be mutually distinguishable tangles in a hypergraph $G$ with $n \geq 1$, and let $\lambda$ be a tie-breaker. Then there is a tree-decomposition $(T, \tau)$ of $G$, where $V(T) = \{t_1, \dots, t_n\}$, with the following properties:*

(i) *For every $e \in E(T)$ and for $1 \leq i \leq n$, if $T_1, T_2$ are the components of $T \setminus e$ and $t_i \in V(T_1)$ then*

$$\left( \bigcup_{t \in V(T_1)} \tau(t), \bigcup_{t \in V(T_2)} \tau(t) \right) \notin \mathscr{T}_i.$$

(ii) *For all $i \neq j$ with $1 \leq i, j \leq n$, let $e$ be the edge of the path of $T$ between $t_i$ and $t_j$ making separations of smallest $\lambda$-order; then these separations are the $(\mathscr{T}_i, \mathscr{T}_j)$- and $(\mathscr{T}_j, \mathscr{T}_i)$-distinctions.*

*Proof.* For $i \neq j$ with $1 \leq i, j \leq n$, there is a $(\mathscr{T}_i, \mathscr{T}_j)$-distinction. Each of these separations is doubly $\lambda$-robust by (10.2), and so by (9.4) no two of them cross. By (9.1) there is a tree-decomposition $(T, \tau)$ of $G$ such that

(i) for $1 \leq i, j \leq n$ with $i \neq j$, a unique edge of $T$ makes the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction

(ii) for every $e \in E(T)$, there exist $i \neq j$ with $1 \leq i, j \leq n$ such that $e$ makes the $(\mathscr{T}_i, \mathscr{T}_j)$- and $(\mathscr{T}_j, \mathscr{T}_i)$-distinctions.

[p. 182] For $1 \leq i \leq n$, we say $t_0 \in V(T)$ is a *home* for $\mathscr{T}_i$ if for every $e \in E(T)$,

$$\left( \bigcup_{t \in V(T_1)} \tau(t), \bigcup_{t \in V(T_2)} \tau(t) \right) \notin \mathscr{T}_i,$$

where $T_1, T_2$ are the components of $T \setminus e$ and $t_0 \in V(T_1)$.

(1) *For $t_0 \in T$ and $1 \leq i < j \leq n$, $t_0$ is not a home for both $\mathscr{T}_i$ and $\mathscr{T}_j$.*

For let $e$ be an edge of $T$ making the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction. Let $T_1, T_2$ be the components of $T \setminus e$, where the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction $(A, B)$ is

$$\left( \bigcup_{t \in V(T_1)} \tau(t), \bigcup_{t \in V(T_2)} \tau(t) \right).$$

Then $(A, B) \in \mathscr{T}_i$ and $(B, A) \in \mathscr{T}_j$, and so if $t_0$ is a home for $\mathscr{T}_i$ then $t_0 \notin V(T_1)$, and if $t_0$ is a home for $\mathscr{T}_j$ then $t_0 \notin V(T_2)$. Since $t_0 \in V(T_1 \cup T_2)$, $t_0$ is not a home for both $\mathscr{T}_i$ and $\mathscr{T}_j$, as required.

For the moment, fix $i$ with $1 \leq i \leq n$. An edge $e \in E(T)$ is *$i$-relevant* if the separations made by $e$ have order less than the order of $\mathscr{T}_i$. Let us direct each $i$-relevant edge $e$ so that

$$\left( \bigcup_{t \in V(T_1)} \tau(t), \bigcup_{t \in V(T_2)} \tau(t) \right) \in \mathscr{T}_i,$$

where $T_1, T_2$ are the components of $T \setminus e$ and $V(T_2)$ contains the head of $e$. We observe that

(2) *$t_0 \in V(T)$ is a home for $\mathscr{T}_i$ if and only if every $i$-relevant edge of $T$ is directed towards $t_0$.*

Let $H_i$ be the set of homes for $\mathscr{T}_i$.

(3) *$H_i \neq \emptyset$ and $H_i$ is the set of vertices of a subtree of $T$.*

The second assertion follows from the first and (2). To show that $H_i \neq \emptyset$, it suffices (by an elementary property of trees) to show that for all $i$-relevant edges $e, e'$ of $T$, if $T_1, T_2$ are the components of $T \setminus e$ with the head of $e$ in $V(T_2)$, and $T_1', T_2'$ are defined similarly, then $V(T_2) \cap V(T_2') \neq \emptyset$. Now

$$\left( \bigcup_{t \in V(T_1)} \tau(t), \bigcup_{t \in V(T_2)} \tau(t) \right) \in \mathscr{T}_i$$

and

$$\left( \bigcup_{t \in V(T_1')} \tau(t), \bigcup_{t \in V(T_2')} \tau(t) \right) \in \mathscr{T}_i,$$

[p. 183] and so $T_2' \not\subseteq T_1$ by the second tangle axiom; thus, $T_2 \cap T_2'$ is non-null, as required.

(4) *If $e \in E(T)$ has ends $x, y \in V(T)$, and $x \in H_i$, $y \notin H_i$, then $e$ is $i$-relevant.*

For since $x \in H_i$ and $y \notin H_i$, some edge of $T$ is directed towards $x$ and not towards $y$. The only possible such edge is $e$, and so $e$ is directed and hence $i$-relevant.

(5) *For $1 \leq i, j \leq n$, and $e \in E(T)$, $e$ makes a separation which distinguishes $\mathscr{T}_i$ from $\mathscr{T}_j$ if and only if $e$ lies on the (unique) minimal path of $T$ between $V(H_i)$ and $V(H_j)$ and is $i$- and $j$-relevant.*

For if $e$ makes a separation which distinguishes $\mathscr{T}_i$ from $\mathscr{T}_j$, this separation has order less than the smaller of the orders of $\mathscr{T}_i, \mathscr{T}_j$, and so $e$ is $i$-relevant and $j$-relevant, and from (2), $e$ lies on the unique minimal $H_i - H_j$ path in $T$. Conversely, if $e$ lies on this path and is $i$- and $j$-relevant, then it makes a separation $(A, B)$ with $(A, B) \in \mathscr{T}_i$ and $(B, A) \in \mathscr{T}_j$, by definition of $H_i$ and $H_j$, as required.

(6) *For $1 \leq i \leq n$, $|H_i| = 1$.*

For suppose that $|H_i| \geq 2$ for some $i$. Choose $t_1, t_2 \in H_i$, distinct and adjacent in $T$ (this is possible by (3)) joined by an edge $e$. Then $e$ is not $i$-relevant. Choose $j, k$ with $j \neq k$ and $1 \leq j, k \leq n$ such that $e$ makes the $(\mathscr{T}_j, \mathscr{T}_k)$-distinction. Let $P$ be the minimal $H_j - H_k$ path in $T$. Then $e \in E(P)$ by (5), and so $j, k \neq i$. Let $f \in E(T)$ make the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction. Then by (5), $f \in E(P)$. Since $f$ is $i$-relevant and $e$ is not, $f$ makes a separation of order (and hence $\lambda$-order) strictly smaller than that of the $(\mathscr{T}_j, \mathscr{T}_k)$-distinction, and by (5) makes a separation of that order which distinguishes $\mathscr{T}_j$ from $\mathscr{T}_k$, a contradiction, as required.

(7) $H_1 \cup \cdots \cup H_n = V(T)$.

For suppose that $t_0 \in V(T) - (H_1 \cup \cdots \cup H_n)$. Since $n \neq 0$, $|V(T)| \geq 2$, and so there is a neighbour of $t_0$ in $T$. Let the edges of $T$ incident with $t_0$ be $e_1, \ldots, e_k$, let $T_p$ be the component of $T \setminus e_p$ not containing $t_0$, and let $T'_p$ be the other component of $T \setminus e_p$ $(1 \leq p \leq k)$. The separations made by $e_1, \ldots, e_k$ are all distinct, since each of them is the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction for some $i, j$, and the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction is made by a unique edge, from our initial choice of the tree-decomposition. Thus we may assume, by the first tie-breaker axiom, that the separations made by $e_1$ have $\lambda$-order strictly more than the separations made by $e_2, \ldots, e_k$. Choose $i, j$ with $i \neq j$ such that

$$\left( \bigcup_{t \in V(T_1)} \tau(t), \bigcup_{t \in V(T'_1)} \tau(t) \right)$$

[p. 184] is the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction. Let $P$ be the minimal $H_i - H_j$ path in $T$. Then $e_1 \in E(P)$, and since $t_0 \notin H_i \cup H_j$, $E(P)$ contains one of $e_2, \ldots, e_k$,

say $e_2$. Now

$$\left(\bigcup_{t\in V(T_2)} \tau(t), \bigcup_{t\in V(T_2')} \tau(t)\right)$$

has $\lambda$-order strictly less than that of the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction and hence has order at most that of the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction. By (5), $e_2$ makes a separation which distinguishes $\mathscr{T}_i$ from $\mathscr{T}_j$, with $\lambda$-order strictly smaller than that of the $(\mathscr{T}_i, \mathscr{T}_j)$-distinction, a contradiction.

Let $H_i = \{t_i\}$ $(1 \le i \le n)$; then the theorem is satisfied. □

We call the tree-decomposition of (10.3) the *standard tree-decomposition* of $G$ relative to $\mathscr{T}_1, \dots, \mathscr{T}_n$.

From (10.3) we deduce a corollary mentioned earlier. We merely sketch the proof since we do not need the result.

(10.4) *In any hypergraph $G$ there are at most $|V(G)|$ maximal tangles.*

*Proof.* Let $\mathscr{T}_1, \dots, \mathscr{T}_n$ be the distinct maximal tangles in $G$, and let $\lambda$ be a tie-breaker. Since they are mutually distinguishable, there is a standard tree-decomposition $(T, \tau)$. Let $e, f \in E(T)$ be distinct, making separations $(A, B)$ and $(C, D)$, say, where $A \subseteq C$ and $D \subseteq B$. If $V(A) = V(C)$ then it follows easily that $A = C$, $B = D$, a contradiction; thus $V(A) \subset V(C)$ and similarly $V(D) \subset V(B)$. From this one can show that $|E(T)| \le |V(G)| - 1$, and hence $n = |V(T)| \le |V(G)|$, as required. We give the details. There are only finitely many tangles in $G$, since each of them is a set of separations of $G$. Two distinct maximal tangles are distinguishable, since if one of them were a subset of the other they would be equal, by maximality. A tie-breaker exists by (9.2), and so if $n \ge 1$, the standard tree-decomposition $(T, \tau)$ exists by (10.3), where $t_i$ corresponds to $\mathscr{T}_i$. Let $\theta_i$ be the order of $\mathscr{T}_i$. For an edge $e$ of $T$ with ends $t_i, t_j$, let $G_i^e$ be the union of $\tau(t)$ over the vertices $t$ of the component of $T \setminus e$ that contains $t_i$, as in the proof of (9.1). By (10.3)(ii), applied to $t_i$ and $t_j$, whose path in $T$ has only the edge $e$, one of the two separations made by $e$ is the $(\mathscr{T}_j, \mathscr{T}_i)$-distinction. It is in $\mathscr{T}_j$ and its reverse is in $\mathscr{T}_i$. It is not $(G_j^e, G_i^e)$, since $(G_i^e, G_j^e) \notin \mathscr{T}_i$ by (10.3)(i). Hence

$$(G_i^e, G_j^e) \in \mathscr{T}_j \quad \text{and} \quad (G_j^e, G_i^e) \in \mathscr{T}_i.$$

We first prove the inclusions $V(A) \subset V(C)$ and $V(D) \subset V(B)$ of the sketch. As in the proof of (9.1), we may name the ends of $e$ as $t_i, t_j$ and those of $f$ as $t_k, t_l$ so that the ends of $f$ are in the component of $T \setminus e$ that contains $t_j$, and the ends of $e$ are in the component of $T \setminus f$ that contains $t_k$. Then $(A, B) = (G_i^e, G_j^e)$ and $(C, D) = (G_k^f, G_l^f)$ satisfy $A \subseteq C$ and $D \subseteq B$, as shown there. Suppose that $V(A) = V(C)$. Then $V(C \cap D) \subseteq V(A \cap B)$ since

$V(D) \subseteq V(B)$, and so $(C, D)$ has order less than $\theta_j$, because $(A, B) \in \mathscr{T}_j$. Since $V(C) \subseteq V(A)$, it follows that $(C, D) \in \mathscr{T}_j$ by (2.9)(ii). But $t_j$ is in the component of $T \setminus f$ that contains $t_k$, and so $(C, D) \notin \mathscr{T}_j$ by (10.3)(i), a contradiction. Thus $V(A) \subset V(C)$. The same argument, applied to $(D, C) \in \mathscr{T}_k$ and $(B, A)$, shows that $V(D) \subset V(B)$.

For the bound on $n$ we show the following.

(1) *If $t_i, t_j$ are adjacent in $T$, then $V(\tau(t_i)) \not\subseteq V(\tau(t_j))$.*

Let $e$ be the edge joining $t_i$ and $t_j$, and put $(A, B) = (G_i^e, G_j^e)$. Then $(B, A) \in \mathscr{T}_i$, and so $(A, B) \notin \mathscr{T}_i$ by (2.1), and $(A, B)$ has order less than $\theta_i$. Suppose that $V(\tau(t_i)) \subseteq V(\tau(t_j))$. Since $\tau(t_i) \subseteq A$ and $\tau(t_j) \subseteq B$, it follows that $V(\tau(t_i)) \subseteq V(A \cap B)$. Let $t_{k_1}, \ldots, t_{k_m}$ be the neighbours of $t_i$ other than $t_j$, let $e_h$ be the edge joining $t_i$ and $t_{k_h}$, and put $(A_h, B_h) = (G_{k_h}^{e_h}, G_i^{e_h})$ $(1 \leq h \leq m)$. Then $(A_h, B_h) \in \mathscr{T}_i$, as shown above. Also $V(A_h \cap B_h) \subseteq V(\tau(t_i))$, by (iii) of the definition of tree-decomposition, since $t_i$ is on the path of $T$ between any vertex of the component of $T \setminus e_h$ that contains $t_{k_h}$ and any vertex of the other component. If $m = 0$, then $t_i$ is the only vertex of its component of $T \setminus e$, so $A = \tau(t_i)$ and $V(A) \subseteq V(B)$. Then $V(B) = V(G)$, contrary to the third axiom, since $(B, A) \in \mathscr{T}_i$. So $m \geq 1$. For $1 \leq h \leq m$, every vertex of $(A_1 \cup \cdots \cup A_h) \cap (B_1 \cap \cdots \cap B_h)$ is in $A_g \cap B_g$ for some $g \leq h$, and so it is in $V(\tau(t_i)) \subseteq V(A \cap B)$. Hence this separation has order less than $\theta_i$, and by (2.2) and induction on $h$ it is in $\mathscr{T}_i$. Put $C = A_1 \cup \cdots \cup A_m$ and $D = B_1 \cap \cdots \cap B_m$, so that $(C, D) \in \mathscr{T}_i$. We claim that $V(D) \subseteq V(B)$. Let $x \in V(D)$, and suppose that $x \notin V(B)$. Then $x \in V(A)$, and so $x \in V(\tau(t))$ for some vertex $t$ of the component of $T \setminus e$ that contains $t_i$. Here $t \neq t_i$ since $V(\tau(t_i)) \subseteq V(B)$, and so the path of $T$ from $t_i$ to $t$ has first edge $e_h$ for some $h$. Since $x \in V(B_h)$, there is a vertex $t'$ of the component of $T \setminus e_h$ that contains $t_i$ with $x \in V(\tau(t'))$. Then $t_i$ is on the path of $T$ between $t'$ and $t$, and so $x \in V(\tau(t_i)) \subseteq V(B)$, a contradiction. Thus $V(D) \subseteq V(B)$, and so $(A, B) \in \mathscr{T}_i$ by (2.9)(i), a contradiction. This proves (1).

If $n = 0$ there is nothing to prove, and if $n = 1$ then $n \leq |V(G)|$ since $\mathscr{T}_1$ has order at least 1 and at most $|V(G)|$. So we may assume that $n \geq 2$. For each vertex $t \neq t_1$ of $T$, let $p(t)$ be the neighbour of $t$ on the path of $T$ from $t$ to $t_1$, let $e_t$ be the edge joining $t$ and $p(t)$, and choose $x(t) \in V(\tau(t)) - V(\tau(p(t)))$, which is possible by (1). Since $n \geq 2$, $t_1$ has a neighbour, and so $V(\tau(t_1)) \neq \emptyset$ by (1). Choose $x(t_1) \in V(\tau(t_1))$. We claim that $x(t) \neq x(t')$ for all distinct $t, t'$. Suppose that $x(t) = x(t') = x$. If $t \neq t_1$ and $t'$ is in the component of $T \setminus e_t$ that contains $p(t)$, then $p(t)$ is on the path of $T$ between $t$ and $t'$, and so $x \in V(\tau(p(t)))$ by (iii), contrary to the choice of $x(t)$. Similarly it is impossible that $t' \neq t_1$ and $t$ is in the component of $T \setminus e_{t'}$ that contains $p(t')$. If $t = t_1$, then $t' \neq t_1$, and $t = t_1$ is

in the component of $T \setminus e_{t'}$ that contains $p(t')$, by the choice of $p(t')$, which we have seen is impossible. So $t \neq t_1$, and similarly $t' \neq t_1$. Hence $t'$ is in the component of $T \setminus e_t$ that contains $t$, and $t$ is in the component of $T \setminus e_{t'}$ that contains $t'$. In particular $e_t \neq e_{t'}$, since otherwise $t' = p(t)$. The component of $T \setminus e_t$ that contains $p(t)$ also contains $t_1$, by the choice of $p(t)$, and it does not contain $t'$. So it does not contain $e_{t'}$, and $p(t)$ and $t_1$ are in the same component of $T \setminus e_{t'}$. Since $t'$ and $t$ are in the same component of $T \setminus e_{t'}$, and $t$ and $p(t)$ are joined by $e_t \neq e_{t'}$, it follows that $t'$ and $t_1$ are in the same component of $T \setminus e_{t'}$. This is contrary to the choice of $p(t')$, which puts $t'$ and $t_1$ in different components of $T \setminus e_{t'}$. This proves our claim. Hence $n = |V(T)| \leq |V(G)|$, and $|E(T)| = n - 1 \leq |V(G)| - 1$.[F5] □

## 11. Structure Relative To A Tangle

Now we come to the last main result of the paper. We have seen in (5.2) that if $G$ has small tangle number, then it has a tree-decomposition of small width. Our problem here is, suppose that $G$ has large tangle number, but relative to each high order tangle the graph has a structure or decomposition of a certain kind $X$, say; what can we infer about the global structure of $G$ from this local knowledge? One might guess that $G$ should have a tree-decomposition into pieces each with structure $X$, but that is false. Nevertheless, it turns out that $G$ has a tree-decomposition into pieces which "almost" have structure $X$, and we need to know this for an application in [6].

[p. 185] A design is a pair $(H, M)$, where $H$ is a hypergraph and $M$ is a set of subsets of $V(H)$. If $(T, \tau)$ is a tree-decomposition of a hypergraph $G$ and $t_0 \in V(T)$, and $t_0$ has neighbours $t_1, \ldots, t_k$ in $T$, then

$$(\tau(t_0), \{V(\tau(t_0) \cap \tau(t_i)) : 1 \leq i \leq k\})$$

is a design, called the *design of* $t_0$ *in* $(T, \tau)$. If $\mathscr{S}$ is a class of designs, a tree-decomposition $(T, \tau)$ is said to be *over* $\mathscr{S}$ if for each $t_0 \in V(T)$, $\mathscr{S}$ contains the design of $t_0$ in $(T, \tau)$.

Let $(H, M)$, $(H', M')$ be designs and let $Z \subseteq V(H')$ be such that (i) $H$ is a subhypergraph of $H'$ and $V(H') - V(H) \subseteq Z$ (ii) every edge of $H'$ is an edge of $H$ (iii) for every $X \in M'$ with $X \neq Z$, $X \cap V(H) \in M$.

($Z$ may or may not be a member of $M'$.) In these circumstances, we say that $(H', M')$ is an *$n$-enlargement* of $(H, M)$ for every integer $n \geq |Z|$. If $\mathscr{S}$

[F5] **Missing proof.** The paper only sketches this proof. The details above are the argument used in the formalization.

is a class of designs, we denote the class of all $n$-enlargements of members of $\mathscr{S}$ by $\mathscr{S}^n$. For any integer $n \geq 0$, we denote by $\mathscr{R}_n$ the class of all designs $(H, M)$ with $|V(H)| \leq n$.

A *location* in a hypergraph $G$ is a set $\{(A_1, B_1), \ldots, (A_k, B_k)\}$ of separations of $G$ such that $A_i \subseteq B_j$ for all distinct $i, j$ with $1 \leq i, j \leq k$. If $\{(A_1, B_1), \ldots, (A_k, B_k)\}$ is a location in $G$, then

$$(G \cap B_1 \cap \cdots \cap B_k; \{V(A_i \cap B_i) : 1 \leq i \leq k\})$$

is a design, which we call the *design of the location*.

Let $\theta \geq 1$ be an integer, and let $\mathscr{S}$ be a class of designs. We say that $\mathscr{S}$ is *$\theta$-pervasive* in a hypergraph $G$ if for every subhypergraph $G'$ of $G$ and every tangle $\mathscr{T}$ in $G'$ of order $\geq \theta$ there is a location $\mathscr{L}$ in $G'$ such that $\mathscr{L} \subseteq \mathscr{T}$ and the design of $\mathscr{L}$ belongs to $\mathscr{S}$. Our object is to deduce information about the global structure of $G$ from the knowledge that a certain class of designs is $\theta$-pervasive. We show

(11.1) *For any $\theta \geq 1$, let $\mathscr{S}$ be a class of designs which is $\theta$-pervasive in a hypergraph $G$; then $G$ has a tree-decomposition over $\mathscr{S}^{3\theta-2} \cup \mathscr{R}_{4\theta-3}$.*

We need the following lemma.

(11.2) *Let $\theta \geq 1$, let $\mathscr{S}$ be $\theta$-pervasive in $G$, and let $Z \subseteq V(G)$ with $|Z| = 3\theta - 2$. Then either* (i) *there is a separation $(A, B)$ of $G$ of order $< \theta$ with*

$$|(Z \cup V(A)) \cap V(B)|, |(Z \cup V(B)) \cap V(A)| \leq 3\theta - 3$$

[p. 186] *or* (ii) *there is a location $\{(A_1, B_1), \ldots, (A_k, B_k)\}$ in $G$, with design in $\mathscr{S}$, such that for $1 \leq i \leq k$,*

$$|Z \cap V(A_i)| \leq |V(A_i \cap B_i)| < \theta.$$

*Proof.* Let $\mathscr{T}$ be the set of all separations $(A, B)$ of $G$ of order $< \theta$ such that $|Z \cap V(A)| \leq |V(A \cap B)|$. Since $|Z| > 3(\theta - 1)$ the second and third tangle axioms hold for $\mathscr{T}$. Suppose the first does not; then there is a separation $(A, B)$ of order $< \theta$ such that $|Z \cap V(A)|, |Z \cap V(B)| > |V(A \cap B)|$. But then

$$\begin{aligned} |(Z \cup V(A)) \cap V(B)| &= |V(A \cap B)| + |Z - V(A)| \\ &< |Z \cap V(A)| + |Z - V(A)| = |Z| = 3\theta - 2 \end{aligned}$$

and similarly $|(Z \cup V(B)) \cap V(A)| \leq 3\theta - 3$, and so (i) holds. We may assume then that $\mathscr{T}$ is a tangle of order $\theta$.

Since $\mathscr{S}$ is $\theta$-pervasive, there is a location $\{(A_1, B_1), \ldots, (A_k, B_k)\} \subseteq \mathscr{T}$ with design in $\mathscr{S}$. Thus for $1 \leq i \leq k$, $|Z \cap V(A_i)| \leq |V(A_i \cap B_i)| < \theta$, and so (ii) holds, as required. $\square$

If $(H, M)$ is a design and $Z \subseteq V(H)$ then $(H, M \cup \{Z\})$ is a design, which we call the *$Z$-extension* of $(H, M)$. In order to prove our main result (11.1) it is convenient for inductive purposes to prove a somewhat strengthened form, the following ((11.1) may be derived from this by setting $Z = \emptyset$).

(11.3) *Let $\mathscr{S}$ be a class of designs, and let $\theta \geq 1$. Let $G$ be a hypergraph such that $\mathscr{S}$ is $\theta$-pervasive in $G$, and let $Z \subseteq V(G)$ with $|Z| \leq 3\theta - 2$. Then there is a tree-decomposition $(T, \tau)$ of $G$ over $\mathscr{S}^{3\theta-2} \cup \mathscr{R}_{4\theta-3}$, such that for some $t_0 \in V(T)$, $Z \subseteq V(\tau(t_0))$ and $\mathscr{S}^{3\theta-2} \cup \mathscr{R}_{4\theta-3}$ contains the $Z$-extension of the design of $t_0$ in $(T, \tau)$.*

*Proof.* Let us remark, first, that from the definition of $\theta$-pervasive, if $\mathscr{S}$ is $\theta$-pervasive in $G$ then it is $\theta$-pervasive in every subhypergraph of $G$. Let $\mathscr{S}' = \mathscr{S}^{3\theta-2} \cup \mathscr{R}_{4\theta-3}$. For fixed $\mathscr{S}, \theta$, we prove that the result holds for all $G, Z$ by induction on $|V(G)|$. Thus, let us assume that it holds for all $G', Z'$ with $|V(G')| < |V(G)|$. First we show that it holds for $G, Z$ if $|Z| = 3\theta - 2$.

Therefore, let $|Z| = 3\theta - 2$. By (11.2), one of the following two cases applies.

*Case 1.* There is a separation $(A_1, A_2)$ of $G$ of order $< \theta$, with

$$|(Z \cup V(A_1)) \cap V(A_2)|, |(Z \cup V(A_2)) \cap V(A_1)| \leq 3\theta - 3.$$

[p. 187] Let $Z_1 = (Z \cup V(A_2)) \cap V(A_1)$, $Z_2 = (Z \cup V(A_1)) \cap V(A_2)$. Then for $i = 1, 2$, $Z_i \subseteq V(A_i)$ and $|Z_i| \leq 3\theta - 3$. Since $|Z_1| < |Z|$ and so $Z \not\subseteq Z_1$, it follows that $V(A_1) \neq V(G)$, and so the result holds for $A_1, Z_1$, and similarly for $A_2, Z_2$ by our inductive hypothesis. Since $\mathscr{S}$ is $\theta$-pervasive in $A_1$ and in $A_2$, it follows that for $i = 1, 2$, there is a tree-decomposition $(T_i, \tau_i)$ of $A_i$ over $\mathscr{S}'$, and there exists $t_i \in V(T_i)$ such that $Z_i \subseteq V(\tau_i(t_i))$ and $\mathscr{S}'$ contains the $Z_i$-extension of the design of $t_i$ in $(T_i, \tau_i)$. We choose $T_1, T_2$ to be disjoint. Take a new vertex $t_0$, and let $T$ be the tree with vertex set $V(T_1) \cup V(T_2) \cup \{t_0\}$, where $T \setminus t_0 = T_1 \cup T_2$ and $t_0$ is adjacent to $t_1, t_2$. Let $\tau(t_0)$ be the hypergraph with vertex set $Z \cup V(A_1 \cap A_2)$ and with no edges, and let $\tau(t) = \tau_i(t)$ if $t \in V(T_i)$ $(i = 1, 2)$. Then $(T, \tau)$ is a tree-decomposition of $G$, as is easily seen. The design of $t_0$ in $(T, \tau)$ is $(\tau(t_0), \{Z_1, Z_2\})$, which is in $\mathscr{R}_{4\theta-3}$, since

$$|V(\tau(t_0))| = |Z \cup V(A_1 \cap A_2)| \leq |Z| + |V(A_1 \cap A_2)| \leq (3\theta - 2) + (\theta - 1),$$

and the $Z$-extension of this design is also in $\mathscr{R}_{4\theta-3}$, for the same reason. For $i = 1, 2$ and each $t \in V(T_i)$, the design of $t$ in $(T, \tau)$ equals the design of

$t$ in $(T_i, \tau_i)$ (or its $Z_i$-extension if $t = t_i$) and so belongs to $\mathscr{S}'$. Hence the theorem holds in this case.

*Case 2.* There is a location $\{(A_1, B_1), \ldots, (A_k, B_k)\}$ in $G$ with design in $\mathscr{S}$, such that for $1 \le i \le k$,

$$|Z \cap V(A_i)| \le |V(A_i \cap B_i)| < \theta.$$

For $1 \le i \le k$, let $Z_i = (Z \cup V(B_i)) \cap V(A_i)$. Then $|Z_i| \le 2(\theta - 1) \le 3\theta - 2$, and $Z_i \subseteq V(A_i)$. Also,

$$|Z \cap V(A_i)| < \theta \le 3\theta - 2 = |Z \cap V(G)|,$$

and so $V(A_i) \ne V(G)$. By our inductive hypothesis, there is a tree-decomposition $(T_i, \tau_i)$ of $A_i$ over $\mathscr{S}'$, and there exists $t_i \in V(T_i)$ such that $Z_i \subseteq V(\tau_i(t_i))$ and $\mathscr{S}'$ contains the $Z_i$-extension of the design of $t_i$ in $(T_i, \tau_i)$. We choose $T_1, \ldots, T_k$ to be disjoint. Take a new vertex $t_0$, and let $T$ be the tree with vertex set $V(T_1) \cup \cdots \cup V(T_k) \cup \{t_0\}$, where $T \setminus t_0 = T_1 \cup \cdots \cup T_k$ and $t_0$ is adjacent to $t_1, \ldots, t_k$. Let $\tau(t_0)$ be the hypergraph with vertex set

$$V(G \cap B_1 \cap B_2 \cap \cdots \cap B_k) \cup Z$$

and with edge set and incidence relation the same as those of $G \cap B_1 \cap B_2 \cap \cdots \cap B_k$. Let $\tau(t) = \tau_i(t)$ if $t \in V(T_i)$ $(1 \le i \le k)$. Then $(T, \tau)$ is a tree-decomposition of $G$, as is easily seen. Let us examine the designs of the vertices of $T$ in $(T, \tau)$. First, let $1 \le i \le k$ and let $t \in V(T_i)$ with $t \ne t_i$.

[p. 188] Then the design of $t$ in $(T, \tau)$ equals the design of $t$ in $(T_i, \tau_i)$, and hence this design belongs to $\mathscr{S}'$. Secondly, let $1 \le i \le k$ and let $t = t_i$; the design of $t$ in $(T, \tau)$ is the $Z_i$-extension of the design of $t$ in $(T_i, \tau_i)$ and hence also belongs to $\mathscr{S}'$. Finally, the design of $t_0$ in $(T, \tau)$ is $(\tau(t_0), \{Z_i : 1 \le i \le k\})$ and its $Z$-extension is $(\tau(t_0), \{Z_i : 1 \le i \le k\} \cup \{Z\})$. But these designs are both $|Z|$-enlargements of

$$(G \cap B_1 \cap \cdots \cap B_k, \{V(A_i \cap B_i) : 1 \le i \le k\}) \in \mathscr{S},$$

and so they both belong to $\mathscr{S}^{3\theta-2} \subseteq \mathscr{S}'$, as required.

Thus, we have proved that the result holds for $G, Z$ when $|Z| = 3\theta - 2$. Now let $Z \subseteq V(G)$ with $|Z| \le 3\theta - 2$. If $|V(G)| < 3\theta - 2$ then $(G, \{Z\}) \in \mathscr{R}_{3\theta-3} \subseteq \mathscr{S}'$, and so the desired tree-decomposition $(T, \tau)$ exists with $T$ a 1-vertex tree. We may assume then that $|V(G)| \ge 3\theta - 2$. Choose $Z' \subseteq V(G)$ with $Z \subseteq Z'$ and $|Z'| = 3\theta - 2$. As we have seen above, the result holds for $G, Z'$, and so there is a tree-decomposition $(T_1, \tau_1)$ of $G$ over $\mathscr{S}'$, such that for some $t_1 \in V(T_1)$, $Z' \subseteq V(\tau_1(t_1))$ and $\mathscr{S}'$ contains the $Z'$-extension of

the design of $t_1$ in $(T_1, \tau_1)$. Take a new vertex $t_0$, and let $T$ be the tree with vertex set $V(T_1) \cup \{t_0\}$, where $T \setminus t_0 = T_1$ and $t_0$ is adjacent to $t_1$. Let $\tau(t_0)$ be the hypergraph with vertex set $Z'$ and no edges, and for $t \in V(T_1)$, let $\tau(t) = \tau_1(t)$. Then $(T, \tau)$ is a tree-decomposition of $G$. For $t \in V(T)$ with $t \neq t_0, t_1$, the design of $t$ in $(T, \tau)$ equals the design of $t$ in $(T_1, \tau_1)$ and hence belongs to $\mathscr{S}'$. The design of $t_1$ in $(T, \tau)$ is the $Z'$-extension of the design of $t_1$ in $(T_1, \tau_1)$ and hence belongs to $\mathscr{S}'$. Finally, the design of $t_0$ in $(T, \tau)$ is $(\tau(t_0), \{Z'\})$, and the $Z$-extension of this is $(\tau(t_0), \{Z, Z'\})$, and both of these belong to $\mathscr{R}_{3\theta-2} \subseteq \mathscr{S}'$. This completes the proof. □

We remark that in essence (11.1) generalizes (5.2). For let $\mathscr{S} = \emptyset$. Then it follows from (11.1) that if $G$ is a hypergraph with no tangle of order $\theta$ (and so $\mathscr{S}$ is $\theta$-pervasive) then $G$ has a tree-decomposition over $\mathscr{R}_{4\theta-3}$, and hence $\omega(G) \leq 4\theta - 4$; in other words, $\omega(G) \leq 4\theta(G)$. Apart from the size of the multiplicative constant, this is the main part of (5.2).

## 12. Tangles and Matroids

Finally, let us discuss some matroidal aspects of tangles. Let $\mathscr{T}$ be a tangle in a hypergraph $G$, of order $\theta$. For $X \subseteq V(G)$, let us define $r(X)$ to be the least order of a separation $(A, B) \in \mathscr{T}$ with $X \subseteq V(A)$, if one exists, and $\theta$ otherwise.

(12.1) *$r$ is the rank function of a matroid on $V(G)$.*

[p. 189] *Proof.* We must check [8] that

(i) $r$ is integral-valued

(ii) for $X \subseteq V(G)$, $0 \leq r(X) \leq |X|$

(iii) for $X \subseteq Y \subseteq V(G)$, $r(X) \leq r(Y)$

(iv) for $X, Y \subseteq V(G)$, $r(X \cup Y) + r(X \cap Y) \leq r(X) + r(Y)$.

(i) and (iii) are clear. For (ii), certainly $r(X) \geq 0$. Since $r(X) \leq \theta$, we may assume that $|X| < \theta$. Let $K$ be the hypergraph with $V(K) = X$, $E(K) = \emptyset$. Since $(G, K) \notin \mathscr{T}$ and has order $< \theta$, it follows that $(K, G) \in \mathscr{T}$, and so

$$r(X) \leq |V(K \cap G)| \leq |X|.$$

This verifies (ii). For (iv), let $X, Y \subseteq V(G)$. Since $r(X \cap Y) \leq r(Y)$ and $r(X \cup Y) \leq \theta$, we may assume that $r(X) < \theta$ and similarly $r(Y) < \theta$. Choose $(A, B) \in \mathscr{T}$ of order $r(X)$ with $X \subseteq V(A)$ and $(C, D) \in \mathscr{T}$ of order $r(Y)$ with $Y \subseteq V(C)$. We claim that $r(X \cap Y)$ is at most the order of $(A \cap C, B \cup D)$; for this is true if $(A \cap C, B \cup D)$ has order $\geq \theta$, and otherwise $(A \cap C, B \cup D) \in \mathscr{T}$, and the claim follows since $X \cap Y \subseteq V(A \cap C)$. Similarly, $r(X \cup Y)$ is at most the order of $(A \cup C, B \cap D)$, by (2.2). Since the sum of

the orders of $(A, B)$ and $(C, D)$ equals the sum of the order of $(A \cap C, B \cup D)$ and $(A \cup C, B \cap D)$, the result follows. $\square$

Thus, given $\mathscr{T}$, $G$ as before, let us say that $X \subseteq V(G)$ is *free* if $|X| \leq \theta$ and there is no $(A, B) \in \mathscr{T}$ of order $< |X|$ with $X \subseteq V(A)$. From (12.1) we deduce

(12.2) *The free sets are the independent sets of a matroid on $V(G)$ with rank function $r$ as in (12.1).*

We shall need (12.2) in a later paper. Incidentally, we do not know which matroids can arise this way, but they are not just the gammoids [8].

Secondly, for the matroid theorist it is a little unnatural to define the order of a separation $(A, B)$ of a graph to be $|V(A \cap B)|$, as we have done. From the viewpoint of matroid theory, a more significant number is the *Tutte-order*, defined to be

$$|V(A \cap B)| + 1 + \kappa(G) - \kappa(A) - \kappa(B),$$

where $\kappa(F)$ denotes the number of components of $F$, for a subgraph $F$ of $G$; for the Tutte-order of a separation $(A, B)$ equals

$$r(E(A)) + r(E(B)) - r(E(G)) + 1,$$

where $r$ is the rank function of the polygon matroid of $G$. One can define both "Tutte-tangles" and "Tutte-branch-width" using Tutte-order instead [p. 190] of order, and the analogue of (4.3) holds. Indeed, this definition of the order of a separation extends to general matroids in the natural way, and again the analogue of (4.3) holds (with essentially the same proof). We suspect, but have not shown, that in a graph, Tutte-tangles and tangles are essentially the same objects. Some evidence for this lies in the fact that, for a connected planar graph, there is a 1-1 correspondence between its tangles and the tangles in a geometric dual [5].

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